Computational Modeling of Belief Structure

A Network-Based Representation of a Focal Belief in a Knowledge Graph

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Declaration of Authorship

I hereby certify that the work presented here is, to the best of my knowledge and belief, original and the result of my own investigations, except as acknowledged, and has not been submitted, either in part or whole, for a degree at this or any other university.

Osnabrück, 27.04.2026

Place, Date

Signature

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To my parents — thank you for everything. None of this would have been possible without your support.

And to my aunt Farida, who has always been an inspiration to me.

Contents

Abstract

Beliefs do not exist in isolation. A focal belief co-occurs with emotional responses and behavioral tendencies in structured ways, yet standard measurement models rarely represent that structure explicitly. This thesis introduces a proof-of-concept workflow that estimates undirected conditional-dependence networks around a focal belief, stores each network as a context-indexed record in a belief knowledge graph, and demonstrates a conditional mean shift analysis that quantifies the implied associational displacement of the focal belief under domain-level conditioning.

The workflow is demonstrated using COSMO survey data from Germany, with perceived COVID-19 severity as the focal belief. Networks were estimated separately across three survey waves and two trust-defined subgroups. Across all six contexts, affective variables were consistently the strongest conditional neighbors of the focal belief, and low-trust networks exhibited higher overall connectivity than high-trust networks. The conditional mean shift analysis confirmed that affect-domain conditioning implied larger focal-belief displacements than behavior-domain conditioning in most contexts. These patterns were stable across multiple robustness checks. The six network instances were encoded in a property-graph knowledge graph enabling structured cross-context retrieval.

The contribution is methodological: the thesis demonstrates a transparent pipeline from survey data to queryable graph-based representation, integrating connectivity-based embeddedness measures, resampling-based stability diagnostics, and a domain-level conditional mean shift framework. All interpretations are descriptive. Estimated edges represent conditional associations and do not imply causal influence or within-person belief change.

The full pipeline is available in Github https://github.com/imarouani/Thesis_Belief_Modeling and deployed at https://belief-modeling.streamlit.app/.

Keywords: focal-belief coupling; psychometric networks; Gaussian graphical model; conditional dependence; embeddedness measures; conditional mean shift; institutional trust; knowledge graph; COSMO; COVID-19.

List of Abbreviations and Symbols

Abbreviations

Abbreviation	Meaning
BKG	Belief Knowledge Graph
CAN	Causal Attitude Network
CMS	Conditional Mean Shift
COSMO	COVID-19 Snapshot Monitoring (German pandemic survey series)
CS-coefficient	Correlation Stability coefficient
EBIC	Extended Bayesian Information Criterion
GGM	Gaussian Graphical Model
GS	Global Strength
MRF	Markov Random Field
RKI	Robert Koch Institute (German federal public health institute)

Notation

Symbol	Meaning
w_{ij}	Estimated partial-correlation edge weight between nodes i and j
$\|w\|$	Absolute edge weight (used in figure captions to indicate edge thickness)
$S(\texttt{SEVERITY})$	Focal strength: sum of absolute weights of all edges incident to the focal node
S_{aff}, S_{beh}	Affect-allocated and behavior-allocated focal strength
P_{aff}	Affect share of focal strength: $S_{\text{aff}}/S(\texttt{SEVERITY})$
GS	Global strength: sum of absolute weights across all edges in a network
γ	EBIC regularization hyperparameter
Σ	Correlation matrix used as input to GGM estimation and conditional mean shift
$\Delta_{\texttt{SEVERITY}}$	Implied conditional displacement of the focal belief in the conditional mean shift analysis (SD units)
c	Conditioning shift magnitude in SD units (set to $c = +1$ throughout)
n, n_{cc}	Sample size; complete-case sample size for a given estimation context
B	Number of bootstrap resamples ($B = 1000$ in this thesis)
r_s	Spearman rank correlation coefficient

1 Introduction and Motivation

1.1 Motivation and problem statement

When people judge a health threat as severe, that judgment does not exist in isolation. It co-occurs with fear and worry, shapes concern about the health system, and predicts whether someone wears a mask or avoids crowded spaces. These components — the belief, the emotions, the behaviors — form a structured configuration, not a loose collection of independent responses (Dalege et al., 2016). Yet standard measurement models rarely make this structure visible. Latent-variable approaches can capture broad covariation and predict outcomes, but they are not designed to show which specific components are directly linked to which others once everything else is accounted for.

1.2 Representational gap

Latent-variable approaches explain observed covariation by positing a shared underlying cause (Borsboom et al., 2003). Under that assumption, measured indicators are conditionally independent given the latent factor, and the model does not represent which specific belief, affect, and behavior components are directly linked to which others. This creates a gap for questions like: *Which components remain associated after controlling for everything else, and how do those direct links differ across subgroups?* The current thesis addresses this gap by representing measured evaluative components as a system whose internal structure can be estimated and inspected (Borsboom & Cramer, 2013).

1.3 Proposed approach: undirected networks and a belief knowledge graph

The approach taken here represents evaluative components as nodes and conditional associations as edges. Empirically, the network is estimated as a regularized Gaussian graphical model (GGM) (Epskamp et al., 2018), resulting in an undirected conditional-dependence structure over a fixed node set. In practical terms, each edge in the network represents the remaining statistical association between two variables after accounting for all other variables in the set — so an edge between fear and severity means these two are linked even after controlling for worry, mask-wearing, and every other measured variable. Focal-belief coupling is summarized using connectivity-based embeddedness measures, which quantify (i) how strongly the focal belief is conditionally linked to other components and (ii) how strongly connected the overall system is within a given estimation condition.

The thesis intentionally begins with a single focal belief embedded in a single local evaluative network. This starting point allows the full estimation and representation pipeline to be specified, demonstrated, and evaluated in a controlled and tractable setting. The belief knowledge graph (BKG) introduced in this thesis is motivated in part by the longer-term goal of supporting *relations across multiple focal-belief networks*, including belief-to-belief connections and, potentially, network-to-network organization. These larger architectural possibilities are future extensions, not something estimated or demonstrated in the current thesis.

For systematic comparison across estimation conditions, each estimated network instance is stored in the BKG. Here, an estimation condition refers to applying the same node set and estimation pipeline to a specific subset of the data. Specifically, estimation conditions are defined as wave $\times$ trust subgroup (low vs. high reported trust in the Robert Koch Institute, RKI). Trust is used only to index estimation conditions and is not included as a node in the belief–affect–behavior network, maintaining a strict separation between context descriptors and the modeled node set. With this operationalization in place, the central question becomes how focal-belief coupling differs across these estimation conditions.

1.4 Research questions and contributions

Research question. How can the structural coupling of a focal belief be estimated using an undirected conditional-dependence network model and represented as context-indexed network instances within a belief knowledge graph to support transparent inspection and comparison across different estimation conditions?

Contributions:

- **Context-indexed belief network estimation pipeline.** This thesis demonstrates that survey responses can be mapped to a family of undirected conditional-dependence networks estimated separately per estimation condition, enabling structured comparison across contexts under a fixed node set and estimation procedure.
- **Operationalization of focal-belief embeddedness.** Focal-belief coupling is quantified using connectivity-based summaries (strength and domain decomposition), treated as descriptive properties of the estimated network structure and evaluated with resampling-based stability diagnostics across four robustness dimensions.
- **Belief knowledge graph architecture.** Estimated network instances are stored together with context descriptors, node-level metrics, edge-level properties, and estimation metadata in a property-graph representation, enabling queryable and auditable cross-context retrieval beyond what standard adjacency-matrix or plot-based outputs support.
- **Conditional mean shift analysis.** A domain-level conditional expectation framework derived from the estimated GGM is used to quantify model-implied associational displacement of the focal belief across node domains and estimation conditions, providing a computational illustration of how embeddedness differences translate to distributional properties.

1.5 Scope and delimitations

All empirical analyses use repeated cross-sectional subgroup data (Betsch et al., 2025). Respondents are not tracked across waves, so it is not possible to study within-person belief change or individual-level dynamics over time. All comparisons are therefore made at the level of group-based estimates defined by the estimation conditions.

The analysis is non-causal. The edges in the estimated networks represent conditional associations in an undirected statistical model and should not be interpreted as causal effects. The conditional mean shift analysis introduced in Chapter 7 follows the same logic: it describes how variables are related within the estimated Gaussian structure, but it does not simulate or predict how beliefs would change under intervention.

As a result, the findings are interpreted as descriptive patterns of conditional dependence within each estimated network. These patterns depend on the chosen node set, preprocessing steps, and estimation procedure. Changing any of these elements would likely lead to different network structures.

The non-causal scope of the thesis is due both to the data and to the modeling approach. Repeated cross-sectional data do not allow tracking individuals over time, and the use of undirected graphical models does not support causal identification. That said, this does not mean that causal analysis is impossible in general. With longitudinal or experimental data, and under additional assumptions, related network-based approaches could be used to study causal processes. Exploring such extensions is beyond the scope of this thesis.

Previous work suggests that belief-related coupling can vary across informational environments, identity relevance, and motivated reasoning (Ecker et al., 2022; Festinger, 1957; Hornsey, 2020; Kahan, 2013). In this thesis, estimation conditions are defined along two dimensions: survey wave (as a proxy for temporal position) and institutional trust, which is treated here as a belief-relevant stratification variable used to define subgroups. This operationalization is intentionally simple and does not cover all possible contextual factors that could shape belief-related coupling.

Estimation settings, such as the EBIC regularization parameter (γ), are not treated as part of the context. Instead, they are recorded separately as estimation metadata. This makes it possible to distinguish between differences in network structure that are due to the estimation conditions and those that arise from modeling choices.

Chapter outline

The thesis proceeds as follows. Chapter 2 develops the theoretical grounding for the approach, covering attitude theory, network psychometrics, the CAN framework, and the motivation for a belief knowledge graph. The remaining chapters operationalize this framework in three layers. The first layer is the data and estimation pipeline: Chapter 3 describes the data source, variable selection, preprocessing, and context operationalization, and Chapter 4 specifies the estimation model, resampling-based stability diagnostics, and embeddedness metrics. The second layer is the representation: Chapter 5 encodes the estimated network instances in a property-graph belief knowledge graph and presents the complete workflow overview. The third layer is the empirical and computational analysis: Chapter 6 reports the descriptive results across all six estimation contexts, and Chapter 7 presents the conditional mean shift analysis along with the code and interactive tool. Chapter 8 interprets the findings, discusses methodological implications, and outlines limitations and future directions.

2 Theoretical Background

To ground the representational choice theoretically, this chapter first introduces the idea that beliefs can be understood as part of broader evaluative systems linking cognitive, affective, and behavioral components. Drawing on attitude theory as a well-established framework for such structured systems, the chapter then motivates a network-psychometric representation as an undirected, descriptive modeling framework.

Here, “belief representation” denotes a network representation of a focal belief embedded in an evaluative system consisting of belief, affect, and behavior. The belief is the analytic focus, while affect and behavior are included to capture the structure of its coupling to other components.

2.1 Beliefs and attitudes as evaluative systems

Classical attitude theories conceptualize attitudes as structured systems integrating cognitive appraisals (beliefs), affective evaluations, and behavioral tendencies. In tripartite formulations (Breckler, 1984; Rosenberg & Hovland, 1960), these components are distinct yet systematically related. From this viewpoint, relations among belief, affect, and behavior are part of the architecture of evaluation. This motivates the representational choice used here: rather than assuming that covariation is fully explained by a latent trait, measured components are treated as potentially interdependent elements of an evaluative configuration.

2.2 Why belief-related evaluations are multi-factorial

Belief-related evaluations in widely debated and often polarized public issues are shaped by multiple interacting influences. Misinformation exposure and correction dynamics can maintain false beliefs through cognitive and social mechanisms even when counterevidence is available (Ecker et al., 2022; Lewandowsky et al., 2012), and repeated exposure increases perceived truth even when the content is implausible

(Hasher et al., 1977; Pennycook et al., 2018). Epistemic trust determines whether evidence sources are treated as legitimate; perceived source credibility has long been recognized as a central determinant of persuasion and message acceptance (Hovland et al., 1953; Pornpitakpan, 2004), and in contested topics, expert communities can be rejected for identity-protective or worldview-consistent reasons (Hornsey, 2020). Social identity and group membership can strongly shape political and evaluative beliefs independently of content (Cohen, 2003; Kahan et al., 2011; Tajfel & Turner, 1986), while motivated reasoning and cognitive dissonance pressures lead individuals to preferentially accept congenial arguments and maintain consistency among cognitions, emotions, and behaviors (Festinger, 1957; Kahan, 2013). Finally, beliefs about threats are typically intertwined with affective responses such as fear and worry, and with protective action tendencies, making belief–affect–behavior configurations emotion-laden systems rather than purely propositional structures (MacKuen et al., 2010; Marcus et al., 2000).

The current thesis does not attempt to model these processes causally. The theoretical point is representational: such influences can co-determine patterns of this coupling, so making component-to-component structure explicit provides a useful descriptive base for later mechanistic work. In practice, the analysis focuses on a small set of nodes capturing (i) a focal belief about COVID-19 severity and (ii) closely related affective responses and protective behavioral tendencies. A belief-relevant stratification variable (institutional trust in the RKI) is used only to stratify the data into estimation subgroups, rather than being included as a node in the core network.

2.3 Context sensitivity as estimation-condition sensitivity

In this thesis, "context sensitivity" refers to the possibility that the conditional-dependence structure among a fixed set of variables differs across estimation conditions. Here, estimation conditions are defined by explicit rules for subsetting the data.

This means that "context" is not treated as an abstract concept, but is operationalized through how the data are partitioned prior to network estimation. Different ways of defining such partitions can reflect different types of contextual indexing. These include belief-relevant factors (e.g., institutional trust), which can be operationalized as subgroup definitions (e.g., trust strata), temporal position (e.g., survey wave), as well as other possible dimensions such as outcome- based or environmental conditions.

In the current thesis, two context dimensions are instantiated: **time (survey wave)** and **belief-relevant subgroup (trust stratum)**. Other forms of contextual indexing are not implemented here and should be understood as possible extensions of the proposed framework.

2.4 Representational limits of latent-variable approaches for this thesis goal

Latent-variable models typically explain observed covariation among indicators by positing an unobserved common cause such that, conditional on that latent variable, the indicators are modeled as approximately independent (Borsboom et al., 2003). This framework is powerful for measurement and prediction, but it does not directly encode which specific measured components are conditionally linked to which others within a defined node set.

The primary object of interest in this thesis is not the strength of a latent attitude dimension, but the explicit conditional-dependence structure among measured belief, affect, and behavior components. From this perspective, residual associations are not nuisance variance but the representational target: they describe how components are linked under the chosen model family. A latent-variable specification would therefore answer a different question than the one pursued here.

2.5 Network psychometrics as a representational alternative

Network psychometrics models psychological constructs as systems of components, represented as nodes and edges (Borsboom & Cramer, 2013; Borsboom et al., 2021). In undirected graphical models, edges correspond to conditional dependence: the association between two variables after controlling for all other variables in the set. Unlike correlation networks, which represent marginal associations, conditional-dependence networks target partial associations and can change when controlling variables are added or removed. This can be useful when the goal is to summarize the structure of direct associations within the modeled set, rather than broad co-movement driven by shared variance. This representation makes structure explicit and supports direct inspection of connectivity patterns.

2.6 The Causal Attitude Network (CAN) framework as theoretical inspiration

2.6.1 Core idea: attitudes as networks of evaluative reactions

The Causal Attitude Network (CAN) model proposes that an attitude is not a single latent cause that generates responses, but a structured system of mutually related evaluative reactions (Dalege et al., 2016, 2019). In the CAN perspective, evaluative reactions include cognitive beliefs (e. g. perceived severity of a threat), affective responses (e. g. fear, worry), and behavioral tendencies (e. g. protective actions). These reactions are represented as nodes in a network, and the attitude is described as emerging from the joint state of this interconnected system (Dalege et al., 2016). For the current thesis, CAN motivates treating the belief–affect–behavior set as an internally structured object and examining how strongly a focal belief is integrated into the surrounding evaluative configuration.

CAN is not limited to a single belief node plus affect and behavior. It is a theory of networks of evaluative reactions, and those reactions can in principle include multiple cognitive or belief-like nodes. Relations between belief nodes are therefore theoretically compatible with CAN already. The current thesis uses a *restricted CAN-inspired implementation* centered on one focal belief. The BKG introduced in this thesis is not what makes belief-to-belief structure theoretically possible; rather, it is what could later help represent, organize, and query larger multi-belief structures if future work extends the network scope beyond a single focal belief.

2.6.2 Consistency pressure and mutual dependence

A central idea in CAN is that evaluative components tend to align with each other. For example, if someone believes that a threat is severe, this typically goes together with feeling worried and supporting protective behaviors, rather than with feeling no concern and rejecting precautions. In this sense, the components are not independent but constrain each other.

Within the CAN framework, the network structure reflects how strongly these dependencies are expressed. This supports using connectivity as a way to describe how tightly the different components are linked, while remaining agnostic about causal direction in the present analysis.

2.6.3 Formalization in CAN: the Ising model for binary evaluative reactions

In its formal implementation, CAN models evaluative reactions using an undirected probabilistic network based on the Ising model (Dalege et al., 2016). In this formulation, each node is typically binary (e. g. reaction absent vs. present; disagree vs. agree after dichotomization). The model specifies that the probability of a full configuration of reactions depends on (i) node-specific thresholds (reflecting baseline endorsement tendencies) and (ii) pairwise interaction parameters (reflecting how strongly two reactions constrain each other). Crucially, the Ising model is an undirected Markov random field: edges encode conditional dependence relations between reactions, given all other reactions in the network (Dalege et al., 2016; Lauritzen, 1996). Thus, even though CAN is a causal theory of mutual influence and consistency, its canonical network representation is an undirected statistical model that captures structured interdependence between evaluative components.

2.6.4 CAN's attitude strength hypothesis and interpretational boundaries

A central claim of CAN is that attitude strength is related to the connectivity of the evaluative network (Dalege et al., 2016). In highly connected networks, evaluative reactions mutually reinforce each other, making the overall attitude more stable and harder to perturb; in less connected networks, reactions are more weakly coupled and the attitude is predicted to be less stable. CAN discusses this in terms of how connectivity supports properties such as stability of evaluative reactions over time, resistance to change, and impact on behavior when behavior nodes are strongly embedded in the evaluative system (Dalege et al., 2016). These are theoretical claims about attitude dynamics.

The current thesis does not test such dynamics. CAN is a causal theory of mutual influence, but the present study is repeated cross-sectional and estimates undirected conditional-dependence networks; estimated edges are therefore interpreted as conditional associations, not causal influences (Borsboom et al., 2021; Epskamp et al., 2018). What is adopted from CAN is its *representational format*: evaluative reactions—cognitive beliefs, affective responses, and behavioral tendencies—are depicted as an interconnected system whose structure can be estimated and inspected, as illustrated schematically in Figure 2.1.

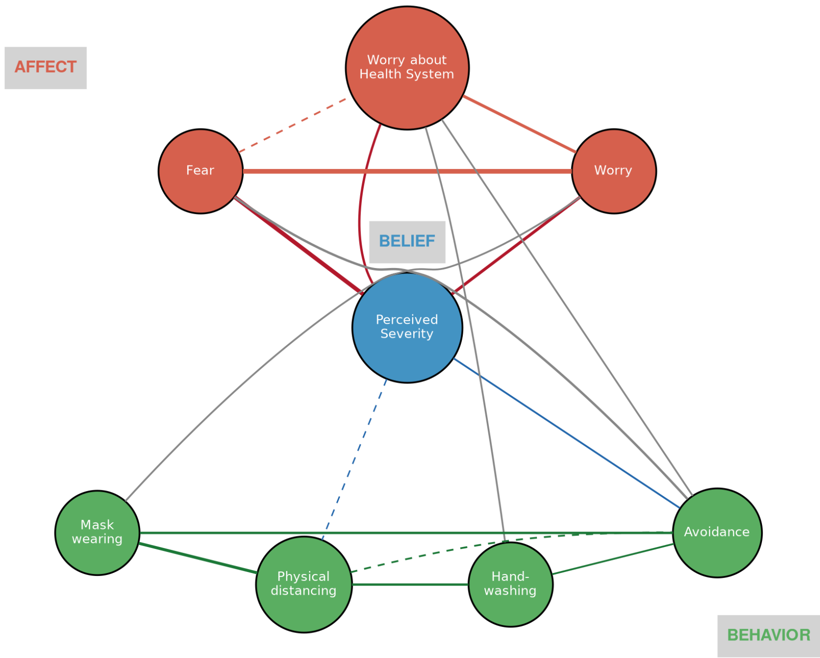

Figure 2.1: Schematic illustration of a CAN-inspired belief–affect–behavior network applied to the current thesis variables. Blue: focal belief (`SEVERITY`); red: affective reactions (`AFF_FEAR`, `AFF_WORRY`, `WORRY_HEALTH_SYSTEM`); green: protective behaviors (`USE2_MASK`, `USE2_SPACE150`, `USE2_HANDWASH20`, `USE2_AVOID`). Edge thickness reflects schematic coupling strength. Dashed edges indicate weaker or less stable associations. Variation in edge appearance is schematic only and does not represent regularization outcomes.

2.7 Statistical framework for estimating undirected conditional-dependence structure

2.7.1 From CAN to a continuous approximation

As discussed previously, CAN is typically formalized using an Ising model for binary evaluative reactions (Dalege et al., 2016). In the present thesis, this formulation is not used directly. The available data consist of Likert-type ordinal survey responses rather than binary indicators, and the design is cross-sectional rather than within-person.

For this reason, the analysis adopts a Gaussian graphical model (GGM) as a continuous approximation to the same representational target: an undirected conditional-dependence structure among belief, affect, and behavior variables. This allows the core idea of CAN—a coupled evaluative system—to be retained, while adapting the statistical model to the properties of the data.

2.7.2 Markov random fields and conditional dependence

Undirected psychometric networks can be formalized as Markov random fields (MRFs) (Lauritzen, 1996). In an MRF, variables are represented as nodes, and an edge indicates conditional dependence: two variables remain statistically associated after controlling for all other variables in the set. If no edge is present, the variables are conditionally independent given the others (within the assumed model family and node set) (Lauritzen, 1996).

This interpretation fixes the meaning of edges as descriptive conditional associations and makes it explicit that all conclusions are conditional on the included variables and modeling assumptions.

2.7.3 Gaussian graphical model (GGM)

A Gaussian graphical model (GGM) is a specific type of MRF for approximately continuous variables, where edges correspond to non-zero partial correlations (equivalently, non-zero elements of the precision matrix) (Epskamp et al., 2018; Lauritzen, 1996). Put simply, if two variables share an edge in the GGM, they are statistically related even after controlling for every other variable in the model; if there is no edge, their association is fully explained by the other variables. In this thesis, the GGM is used to estimate the conditional-dependence structure among the selected belief, affect, and behavior variables.

For the main analyses, the ordinal items are treated as numeric within a standard GGM workflow. This is a modeling approximation rather than a claim that the variables are truly continuous. While several variables exhibit skewed distributions, treating ordinal variables with multiple response categories as approximately continuous is a common pragmatic choice in applied network modeling (Rhemtulla et al., 2012). The implications of this approximation are considered in the robustness checks (Section 6.8).

2.8 From CAN to operational embeddedness measures

Within CAN, belief, affect, and behavior indicators are conceptualized as a coupled evaluative system, and connectivity is interpreted as carrying information about how integrated a focal belief is within that system (Dalege et al., 2016). In the current thesis, this idea is translated into an operational definition: structural embeddedness is defined as the degree to which a focal belief is integrated into a belief–affect–behavior network through the number and strength of its conditional associations.

This definition is descriptive and depends on the model used. Embeddedness is treated as a property of the estimated undirected conditional-dependence network, meaning that it reflects the observed pattern of associations under the selected node set and estimation procedure, rather than a claim about causal influence, identity centrality, or within-person stability over time. The thesis operationalizes embeddedness at two levels:

- **Node-level embeddedness (focal strength):** how strongly the focal belief is connected to the rest of the network, measured as the sum of the absolute edge weights on all edges touching the focal node. A higher value means the severity belief is more tightly coupled to the surrounding variables.
- **Network-level embeddedness (global strength):** how tightly connected the entire network is, measured as the sum of all absolute edge weights. A higher value means the whole system of belief, affect, and behavior variables is more densely interconnected.

These measures provide a compact way to compare how focal-belief coupling and overall system connectivity differ across estimation conditions, while remaining within a strictly associational interpretation. Strength centrality as used here corresponds to weighted degree centrality in the general network science literature (Newman, 2010), selected for its direct interpretability within the CAN framework. Throughout the thesis, *focal-belief coupling* refers to the conceptual phenomenon — the degree to which a belief is structurally tied to its neighbors — while *focal embeddedness* and its metric operationalization (focal strength, global strength) refer to the specific quantities used to measure it. Formal definitions are given in Chapter 4.

2.8.1 Descriptive comparison and its limits

Estimated networks are compared across estimation conditions descriptively rather than inferentially. Formal statistical comparison of network structures (e. g. testing whether specific edges differ significantly across groups or waves) requires additional methodology such as the Network Comparison Test or multiple-group estimation (van Borkulo et al., 2023). Such inferential comparison is beyond the present scope, which focuses on demonstrating a representation and storage workflow. The bootstrap summaries introduced in Chapter 4 provide partial information about sampling variability and should be consulted when interpreting cross-context differences in embeddedness metrics.

2.9 Why a belief knowledge graph (BKG)?

2.9.1 Knowledge graphs and their role in the present work

A knowledge graph (KG) is a graph-structured representation in which entities are represented as nodes and relations between them as edges, enriched with type information and properties (Hogan et al., 2021). Because it is graph-structured, a KG constitutes a kind of network in the broad mathematical sense, but it is *not the same type of object* as the psychometric network. The psychometric network is an *estimated statistical model of conditional dependence*; the BKG is a *representation, storage, and retrieval structure* for estimated instances, metadata, and typed relations. These are functionally distinct objects that serve different purposes within the workflow.

Here, the BKG is used for a narrow purpose: to represent estimated belief-network outputs as explicit, queryable objects that remain linked to provenance and estimation settings across many network instances.

Two common implementation families are Resource Description Framework (RDF) graphs (rooted in the Semantic Web) and property graphs. A property-graph representation (Neo4j-oriented) was chosen because the objects to be stored include multiple estimated network instances whose nodes and edges carry numerical attributes (e. g. edge weights, uncertainty summaries, and embeddedness measures) that are instance-scoped. A graph-based representation is especially suitable here because the object being stored is already relational, and because the use case requires typed relations, provenance, context linkage, instance-specific metrics, and queryable traversal across many stored instances.

2.9.2 Network model versus knowledge graph: two different objects

The statistical network estimated here (a regularized partial-correlation GGM) is a model output: an estimated conditional-dependence structure over a fixed node set, under a specific preprocessing pipeline and tuning procedure, for a particular estimation condition. By contrast, the belief knowledge graph (BKG) is a representation and storage layer that turns these model outputs into explicit, queryable objects.

2.9.3 The core problem: estimation outputs do not scale as a representation standard

A network plot or adjacency matrix is useful for a single instance but lacks built-in support for multi-instance storage, explicit context indexing, and structured cross-instance comparison. Once estimation is repeated across multiple conditions—as in this thesis—standard outputs provide no principled mechanism for linking each network to the context and estimation settings that produced it, or for querying and comparing structural properties across instances without manual inspection. A BKG addresses this limitation by providing a structured representation in which network instances, context descriptors, and derived measures can be stored and queried in a unified way.

2.9.4 Design goal and schema principle of the BKG

The BKG is designed to represent each estimated belief–affect–behavior network instance as a standardized, queryable object that supports inspection and comparison across estimation conditions. Specifically, the BKG stores (i) network instances, (ii) variables as stable identities[1], (iii) context descriptors that define the data subset,

[1] Each variable (e.g., SEVERITY, AFF_FEAR) is represented as the same entity across all network instances, allowing nodes to be linked across different estimation conditions rather than treated as context-specific.

(iv) edges and node metrics (including uncertainty summaries), and (v) estimator metadata and provenance.

A key schema principle is the separation between global variable identity and context-specific instantiation. Context-specific quantities (e. g. centralities, descriptives, and edge weights) are properties of a variable within a particular network instance and must not be stored as global truths. The schema therefore represents instance-specific nodes linked to a global `Variable` and to a specific `NetworkInstance`, ensuring that queries remain semantically correct and context-aware. This design also directly supports the possibility of future multi-network extensions: because variable identity is represented stably and independently of any particular network instance, the same variable can recur across multiple local networks with different structural roles—appearing as a response tendency in one estimated network while functioning differently in another—without creating ambiguity in the stored representation.

The advantage of the BKG lies not only in storing components, but in storing *different kinds of relations explicitly.* The schema distinguishes between:

- **Stable identity relations** (`INSTANCE_OF`): linking a context-specific node instantiation to its global variable identity,
- **Instance membership relations** (`HAS_NODE_INSTANCE`, `HAS_EDGE_INSTANCE`): linking node and edge instances to the `NetworkInstance` they belong to,
- **Context-indexing relations** (`HAS_CONTEXT`, `HAS_GAMMA_SPEC`): linking a network instance to its estimation condition and regularization settings,
- **Endpoint-linking relations** (`FROM`, `TO`): linking an `EdgeInstance` to the two node instances it connects (without implying directionality in the underlying statistical model), and
- **Estimation metadata and provenance properties**: stored as attributes on `EdgeInstance` nodes (bootstrap statistics, sample size, preprocessing decisions).

The BKG stores the estimated quantitative output of each network instance at the appropriate instance scope. Edge weights and uncertainty summaries are stored as properties of `EdgeInstance` nodes, while node-level quantities such as centrality or descriptive summaries are stored on node instances within a given `NetworkInstance`. This ensures that estimated relations between variables are represented as properties of a specific estimation instance, rather than as general relations between variables.

2.9.5 Keeping interpretation tied to context

Finally, the BKG helps keep interpretation tied to its estimation condition and uncertainty. When a query retrieves, for example, "the strongest neighbors of `SEVERITY` in a given estimation condition," the result is conditional on the chosen node set, preprocessing decisions, estimation procedure and tuning, the context definition, and sampling variability (captured by uncertainty summaries). The BKG makes these dependencies explicit by storing results together with their context descriptors and estimator metadata.

3 Methods

3.1 Data source

All empirical analyses use data from the COVID-19 Snapshot Monitoring (COSMO) project in Germany (Betsch et al., 2024, 2025). COSMO is a repeated cross-sectional online survey series measuring psychosocial variables relevant to the COVID-19 pandemic, including risk perceptions, affect, institutional trust, and protective behaviors. All analyses are secondary analyses of anonymized survey data.

3.2 Wave selection

Waves were selected using a data-driven availability audit. The aim was to identify a consecutive set of waves in which all variables required for the fixed node set and the stratification variable `TRUST_RKI` were available with sufficient data quality after applying the preprocessing rules described in Section 3.4.

The selection was done in two steps. First, each wave was screened for basic availability. Only waves with all required variables and acceptable levels of non-missing data were kept as candidates. Second, the remaining waves were scored based on four criteria: (i) the complete-case sample size, (ii) variability in behavioral responses (measured via Shannon entropy across the `USE2_*` variables), (iii) the average correlation between behavioral variables (as a proxy for usable signal in network estimation), and (iv) an overall data-quality flag.

Of the 47 waves that passed the availability threshold (requiring $\geq 90\%$ non-missing values on all required variables), all possible consecutive three-wave blocks were scored using the mean and minimum composite score across the block. The block consisting of waves 55, 56, and 57 ranked highest on both criteria, indicating consistently good data quality and sufficient variability across all three waves.

The main analyses therefore use waves 55–57 (indexed by `TIME`), which form a consecutive segment in which the full node set is available and protective behaviors are measured as ordinal frequency items (USE2_MASK, USE2_SPACE150, USE2_HANDWASH20, USE2_AVOID).

3.3 Measures

The node set contains one focal belief variable, affect variables, and protective-behavior frequency variables. Variable selection was constrained by two criteria: conceptual alignment with the three CAN domains (belief, affect, behavior) and consistent availability across all three selected waves. Among the variables meeting these criteria, the selected items were those most directly capturing the focal belief about COVID-19 severity, the affective responses most closely tied to threat appraisal (fear, worry, and health system concern), and the primary protective behaviors measured in COSMO using a consistent ordinal format across waves 55–57.

- **Focal belief:** SEVERITY (perceived COVID-19 severity; 1–7).
- **Affect:** AFF_FEAR, AFF_WORRY, WORRY_HEALTH_SYSTEM (all 1–7).
- **Behavior:** USE2_MASK, USE2_SPACE150, USE2_HANDWASH20, USE2_AVOID (1–6 with a non-substantive first category; see Section 3.4). After recoding category 1 to missing, the effective ordinal range for behavior variables is 2–6, comprising five substantive response categories.

Institutional trust is used to define estimation contexts:

- **Context variable (not a node):** TRUST_RKI (trust in the Robert Koch Institute, RKI; effective range 3–9 after recoding non-substantive categories, where higher values indicate greater institutional trust).

Trust is not included as a node in the belief–affect–behavior network because it serves as a context descriptor for subsetting rather than as part of the modeled evaluative configuration. The unit of analysis is a *network instance* defined by wave and trust stratum (low vs. high; Section 3.5). For each wave, two network instances are constructed from the corresponding respondent subsets. COSMO is repeated cross-sectional; respondents are not tracked across waves.

3.4 Preprocessing and instance construction

Preprocessing is applied identically across waves and trust strata. The goal is comparability across network instances by holding the node set and coding rules constant while allowing the respondent subset to vary by context.

3.4.1 Coding rules

Non-substantive categories. For `USE2_MASK`, `USE2_SPACE150`, `USE2_HANDWASH20`, `USE2_AVOID`, response category 1 (*Trifft nicht zu*) is recoded to missing (`NA`); categories 2–6 are retained as ordered responses, yielding an effective five-point ordinal scale. For `TRUST_RKI`, categories 1 (*keine Angabe möglich*) and 2 (non-substantive; the substantive response range begins at 3) are recoded to `NA`, yielding an effective range of 3–9.

Affective direction alignment. `AFF_FEAR` and `AFF_WORRY` are reverse-coded so that higher values correspond to higher fear and higher worry, respectively. `WORRY_HEALTH_SYSTEM` is left unchanged because higher values already indicate more worry.

3.4.2 Sample construction and exported datasets

Network estimation is based on complete cases within each context (listwise deletion). After filtering to a given wave and trust stratum, only respondents with observed values on all belief–affect–behavior variables are retained. The context variable `TRUST_RKI` is used only to define the strata and is not required beyond this step.

For each context, the resulting dataset is exported and used directly for network estimation. The corresponding sample size (n_{cc}) is recorded for each network instance and reported alongside the results. Across the six network instances, complete-case sample sizes ranged from $n = 309$ to $n = 407$, with a mean of approximately 360; full per-context values are reported in Table 6.1.

Complete-case analysis implicitly assumes that the data are at least missing at random (MAR), meaning that the probability of missingness may depend on observed variables but not on the missing values themselves. This assumption may not fully hold here: recoding response category 1 (*Trifft nicht zu*) to `NA` for the behavioral items introduces structured missingness that may be systematically associated with the underlying behavioral tendency being measured — that is, the missingness is likely informative. The extent to which this violates the MAR assumption is not testable within the present design and represents a limitation of the complete-case approach (see Section 8.5).

3.5 Context operationalization

Trust groups are defined within each wave using terciles of the non-missing `TRUST_RKI` values. Respondents below the lower tercile are assigned to the low-trust group, and respondents above the upper tercile to the high-trust group. The middle tercile is excluded from the analysis. Because `TRUST_RKI` is ordinal and includes tied values, group sizes do not necessarily correspond to exact thirds; sample sizes are therefore reported for each network instance. The exact tercile cut-points for each wave are provided in Appendix B (Table 2).

This design has two implications. First, excluding the middle tercile means that the analyzed sample represents only a subset of respondents in each wave. The results therefore describe the conditional dependence structure of individuals at the lower and upper ends of the trust distribution and do not necessarily generalize to the full sample. Second, tercile boundaries are computed separately for each wave. As a result, group membership is defined relative to the distribution within that wave, and comparisons across waves refer to the high- and low-trust groups at each time point, not to the same individuals over time.

After the estimation procedures and derived metrics are specified in Chapter 4, the resulting network instances are encoded in a property-graph belief knowledge graph (BKG); this representation layer is described in Chapter 5, which also presents the complete workflow overview.

4 Estimation Model and Derived Metrics

This chapter specifies how each belief network is estimated and what quantities are derived from it. Section 4.1 describes the estimation model — a regularized Gaussian graphical model selected via the extended Bayesian information criterion. Section 4.2 covers the resampling-based diagnostics used to assess edge-level and metric-level stability. Section 4.3 defines the connectivity-based embeddedness measures that operationalize focal-belief coupling. Finally, Section 4.4 summarizes the quantities reported per network instance in the Results chapter. All empirical values — including bootstrap distributions, CS-coefficients, and context-specific stability summaries — are reported in Chapter 6.

4.1 Network estimation: GGM via EBICglasso

Network structure is estimated as a regularized Gaussian graphical model (GGM) using the EBIC-based graphical lasso procedure (EBICglasso) (Epskamp et al., 2018; Foygel & Drton, 2010; Friedman et al., 2008). For each network instance, a correlation matrix is computed from the context-specific complete-case data and passed as input to the estimation routine. The procedure identifies which pairs of variables are directly associated after controlling for all other variables, while applying a penalty that shrinks weak associations to exactly zero. This produces a sparse network in which only the strongest conditional associations survive — a useful property when the goal is to identify the core structure rather than model every possible link. In the resulting network, an edge between two nodes is present whenever their estimated partial correlation is non-zero. The absence of an edge means that those two nodes are estimated to be conditionally independent given all other nodes in the network.

Three estimation specifications are used in this thesis, complemented by a correlation-input sensitivity check. Together, these four robustness dimensions assess whether the structural patterns observed under the main specification hold when (i) the degree of regularization is increased, (ii) an additional significance filter is applied, or (iii) the correlation input is changed from Pearson to Spearman. The main specification uses the EBIC hyperparameter $\gamma = 0.5$. The three robustness checks are: a stronger-penalization sensitivity specification with $\gamma = 0.75$; a thresholded variant of the main specification in which weak edges are removed after estimation (described in Section 4.1.3); and a Spearman rank-correlation re-estimation of the main specification (described in Section 4.1.1). Throughout the Results chapter, outputs are labeled by specification (main, sensitivity, thresholded, or Spearman) so that the role of each modeling choice is always explicit.

4.1.1 Ordinal items and the Gaussian approximation

All node variables are ordinal Likert-type items. The baseline estimation pipeline treats them as continuous and uses Pearson correlations as input to the GGM. This is a standard approximation in the network psychometrics literature: for items with five or more ordered response categories and moderate skew, Pearson-based estimation has been shown to perform acceptably (Rhemtulla et al., 2012).

The approximation is not without risk. Pearson correlations can underestimate associations when item distributions are skewed or exhibit strong ceiling or floor effects (Flora & Curran, 2004). In contexts where response distributions are compressed near the ceiling — for example, behavioral items in subgroups with high baseline adoption — the resulting partial correlations may be attenuated, producing sparser networks and lower edge weights. Such attenuation would be difficult to distinguish from genuine conditional independence within the present design.

To assess whether the structural findings depend on this correlation input, the estimation pipeline was also run using Spearman rank correlations. Unlike Pearson correlations, Spearman correlations operate on ranks rather than raw values and are therefore robust to monotone distortions, ceiling and floor effects, and departures from linearity (Flora & Curran, 2004). This makes them a natural sensitivity check for ordinal data: if the key structural patterns — in particular, the composition and ranking of the focal neighborhood — are preserved under Spearman input, the findings are unlikely to be driven by the Pearson approximation. The Spearman robustness check was run for all six main-specification contexts; results are reported in Section 6.8. Polychoric correlations under a latent-Gaussian assumption (Holgado-Tello et al., 2010; Rhemtulla et al., 2012) represent a further alternative not pursued in the present analyses and noted as a possible extension in Section 8.5.

4.1.2 EBIC model selection and the regularization parameter

Model selection follows the extended Bayesian information criterion for sparse Gaussian graphical models (Foygel & Drton, 2010). The EBIC hyperparameter γ controls how strongly the criterion penalizes model complexity: higher values of γ favor sparser networks. The main specification uses $\gamma = 0.5$, a value widely used in the network psychometrics literature. The sensitivity specification uses $\gamma = 0.75$, which imposes stronger penalization and tends to retain only the most robust edges. Comparing the two specifications reveals which structural features persist under increased regularization and which are sensitive to the penalization level.

4.1.3 Thresholded EBICglasso variant

In addition to increasing γ, a second robustness check applies a post-selection significance filter to the main specification. Specifically, partial correlations that do not reach significance under a Fisher z-test approximation ($p > 0.05$) are set to zero after the EBICglasso selection step. This thresholded variant is implemented via the `threshold = TRUE` argument in `bootnet::estimateNetwork` (Epskamp et al., 2012, 2018).

The three robustness checks serve complementary purposes. The $\gamma = 0.75$ specification increases the regularization strength during model selection. The thresholded $\gamma = 0.5$ specification retains the same model-selection criterion but removes edges that fail a significance screen after selection. The Spearman re-estimation changes the correlation input while holding the regularization settings fixed, testing whether the structural findings are sensitive to the Pearson approximation for ordinal data. Together, these checks vary three independent dimensions of the estimation procedure: regularization strength, post-selection filtering, and correlation input.

4.2 Resampling-based stability diagnostics

Regularized network estimates are sensitive to sampling variability: a different random sample from the same population could yield a somewhat different edge structure. To characterize this variability, two complementary resampling procedures are applied to each network instance. Edge-level bootstrap summaries assess how stable individual edges are across resamples. The CS-coefficient assesses how stable the rank ordering of node-level strength centrality is when cases are progressively dropped. Both procedures are described below; the resulting empirical values are reported in Chapter 6.

4.2.1 Edge-level bootstrap summaries

Edge-weight variability is assessed using nonparametric bootstrapping within each network instance (Epskamp et al., 2018). The logic is simple: if we drew a different random sample from the same population, would the same edges appear? To test this, the procedure repeatedly resamples the data with replacement ($B = 1000$ times), re-estimates the network each time, and records which edges appear and how strong they are. For each edge, the bootstrap distribution is summarized by its mean, standard deviation, percentile interval (2.5th and 97.5th percentiles), and the proportion of resamples in which the edge was non-zero:

$$\text{prop\_nonzero} = \frac{1}{B}\sum_{b=1}^{B} \mathbb{1}\left(w_{ij}^{(b)} \neq 0\right),$$

where $w_{ij}^{(b)}$ is the estimated edge weight in resample b. A `prop_nonzero` of 0.95 means the edge appeared in 95% of resamples — it is reliably present. A value of 0.30 means it appeared in only 30% of resamples and may be an artifact of the particular sample.

For descriptive convenience, edges are classified into three stability categories based on `prop_nonzero`: *stable* (≥ 0.80), *moderate* (0.50–0.80), or *unstable* (< 0.50). These thresholds are descriptive heuristics, not formal inferential cutoffs. Cross-context comparisons prioritize stable and moderate edges, while acknowledging that all three categories reflect exploratory assessment within a regularized framework rather than classical hypothesis testing (see Section 8.5).

4.2.2 Bootstrap uncertainty of focal strength

The edge-level bootstrap characterizes stability at the level of individual edges. However, the primary interpretive quantity in this thesis is the focal strength $S(\texttt{SEVERITY})$, which aggregates over all edges incident on the focal node. To assess whether focal strength is robust to sampling variability, the metric is recomputed for each of the $B = 1000$ bootstrap resamples per context and summarized as a percentile interval.

An important methodological note applies to the aggregation procedure used in the present analyses. The interval bounds reported in the Results chapter were computed by summing edge-level percentile values across focal edges, rather than by computing the percentile of the summed focal strength across bootstrap replicates. The sum of marginal quantiles is not the quantile of the sum; the two coincide only when the summands are monotonically co-varying. This aggregation method can produce lower bounds below zero even though $S(\texttt{SEVERITY})$ is non-negative by definition (as a sum of absolute values). All focal-strength bootstrap summaries should therefore be treated as conservative bounding heuristics rather than proper percentile intervals. Computing $S(\texttt{SEVERITY})$ within each replicate and taking quantiles of those values directly would yield tighter, properly bounded intervals. This correction is noted as a methodological refinement for future work.

4.2.3 Case-dropping stability (CS-coefficient)

While the bootstrap summaries above assess variability in edge weights and focal strength, they do not address whether the *rank ordering* of nodes by centrality is stable. The CS-coefficient addresses this question. It quantifies the largest proportion c of cases that can be dropped from the sample while maintaining a mean correlation of at least 0.70 between the original and subsample strength centrality rankings across 95% of subsamples (Epskamp et al., 2018). Values above 0.50 are conventionally considered acceptable. The CS-coefficient is estimated using `bootnet::bootnet( ..., type = "case")` with $B = 1000$ case-dropping resamples per context, using the same deterministic seed scheme as the edge-level analyses.

4.3 Connectivity-based embeddedness measures

The embeddedness of the focal belief within the estimated network is operationalized using three connectivity-based measures. All metrics are computed per estimated network instance and use absolute edge weights to quantify coupling magnitude.

4.3.1 Focal strength

Let w_{ij} denote the estimated edge weight between nodes i and j. The focal strength quantifies how strongly the focal belief is conditionally coupled to the rest of the network:

$$S(\texttt{SEVERITY}) \;=\; \sum_{j \in \mathcal{N}(\texttt{SEVERITY})} |w_{\texttt{SEVERITY},j}|\,, \tag{4.1}$$

where $\mathcal{N}(\texttt{SEVERITY})$ is the set of neighbors of `SEVERITY` — that is, all nodes that share a non-zero edge with the focal node.

4.3.2 Domain-allocated focal strength

Because the network contains both affect and behavior nodes, it is useful to decompose the focal strength by domain. Let $\mathcal{A}$ denote the set of affect nodes and $\mathcal{B}$ the set of behavior nodes. The domain-specific components of focal strength are:

$$S_{\text{aff}}(\texttt{SEVERITY}) = \sum_{j \in \mathcal{A}} |w_{\texttt{SEVERITY},j}| \,, \tag{4.2}$$

$$S_{\text{beh}}(\texttt{SEVERITY}) = \sum_{j \in \mathcal{B}} |w_{\texttt{SEVERITY},j}| \,. \tag{4.3}$$

The affect share expresses the proportion of focal coupling that is directed toward affect nodes:

$$P_{\text{aff}}(\texttt{SEVERITY}) = \frac{S_{\text{aff}}(\texttt{SEVERITY})}{S(\texttt{SEVERITY})}, \tag{4.4}$$

with $P_{\text{beh}}(\texttt{SEVERITY}) = 1 - P_{\text{aff}}(\texttt{SEVERITY})$. This identity holds by construction of the node set: since the neighborhood of SEVERITY within the chosen node set contains only affect and behavior nodes, the domain decomposition is exhaustive. This identity would not generalize to a richer node set that includes additional domains (for example, cognitive or social belief nodes). This decomposition indicates whether the focal belief is more strongly coupled to affective responses or to behavioral tendencies in a given estimation context.

4.3.3 Global strength

Network-level connectivity is summarized by global strength, defined as the sum of absolute edge weights across all unique edges in the network:

$$GS = \sum_{i<j} |w_{ij}| \,. \tag{4.5}$$

Global strength provides a measure of overall network density in terms of coupling magnitude. It serves as a baseline against which changes in focal strength can be contextualized: a drop in focal strength is more interpretable when one knows whether the network as a whole became sparser or whether the change was specific to the focal node.

An important caveat applies to cross-context comparisons of global strength (and, by extension, of any summed absolute-weight metric). EBICglasso selects different sparsity levels depending on sample size, correlation structure, and the regularization path. This means that the number and magnitude of retained edges can differ across contexts for reasons related to estimation rather than to the underlying data-generating process. All cross-context comparisons of these metrics should be interpreted with this caveat in mind (see also Section 8.5).

4.4 Quantities reported per network instance

For each network instance, the Results chapter reports the following: (i) the focal neighborhood, presented as an edge list ranked by $|w_{\texttt{SEVERITY},j}|$ and accompanied by bootstrap summaries (including `prop_nonzero`); (ii) the connectivity metrics $S(\texttt{SEVERITY})$, GS, and the domain decomposition quantities S_{aff}, S_{beh}, and P_{aff}; (iii) the bootstrap distribution of $S(\texttt{SEVERITY})$ per context; and (iv) the CS-coefficient for strength centrality per context; and (v) a Pearson-vs.-Spearman comparison of focal neighborhood composition and embeddedness metrics. Together, these quantities and robustness checks provide the basis for structured comparison of focal-belief embeddedness across estimation conditions.

5 Belief Knowledge Graph Representation of Estimated Networks

This chapter describes how each estimated belief network — together with its context, estimation settings, and node- and edge-level quantities — is encoded in a property-graph belief knowledge graph (BKG). The purpose of this representation layer is not to modify the statistical model defined in Chapter 4 but to organize its outputs in a structured and queryable form. The chapter ends with a complete workflow overview that situates the BKG within the full pipeline from raw COSMO data to the conditional mean shift analysis presented in Chapter 7.

Conceptually, the graph records one estimated network as a `NetworkInstance` (see Figure 5.1). Each `NetworkInstance` is linked to the context in which the network was estimated and contains the node and edge objects that represent the CAN-inspired belief–affect–behavior configuration of that network.

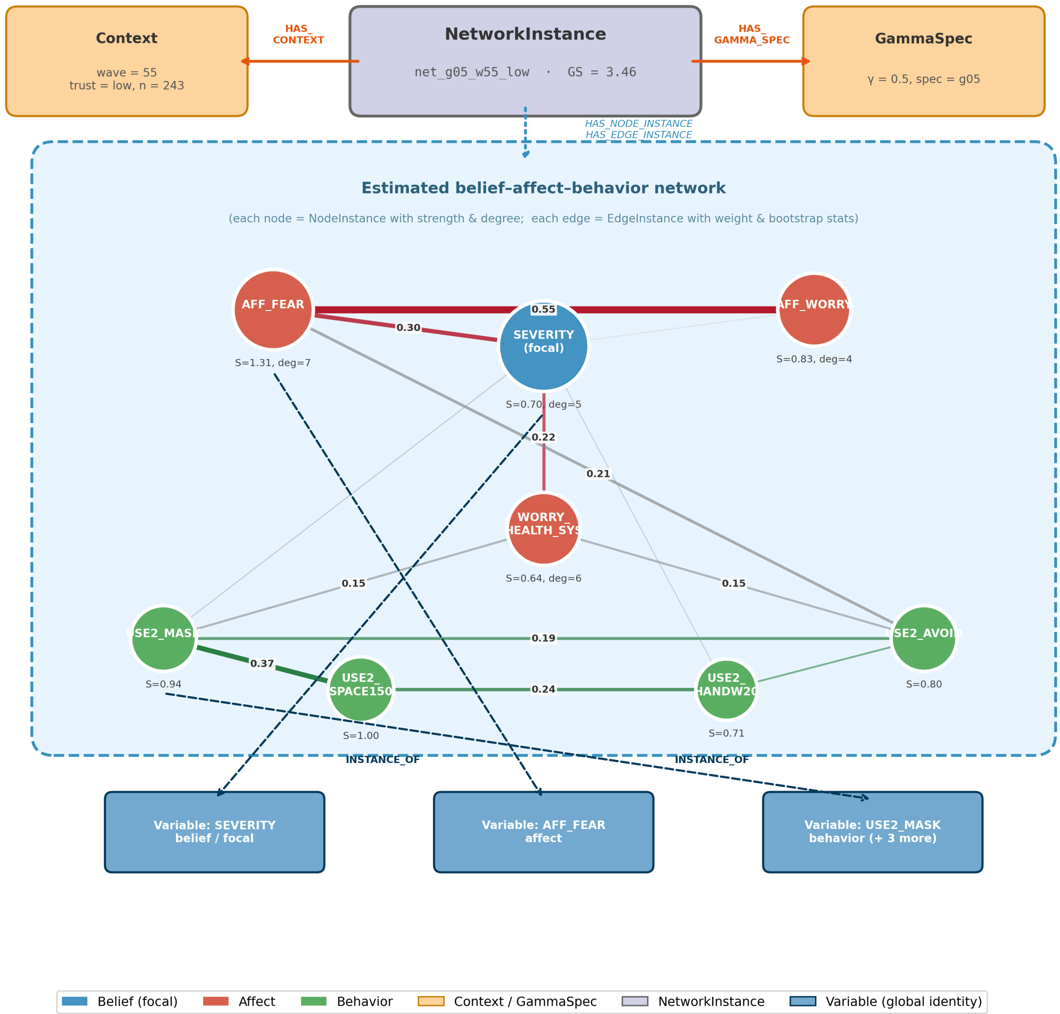

Figure 5.1: Integrated view of a single BKG network instance. **Top:** the `NetworkInstance` node links to its `Context` (wave, trust stratum, sample size); estimation metadata (γ parameter, correlation input) is stored as properties on the `NetworkInstance` node, alongside global strength (GS, the sum of all absolute edge weights). **Middle (dashed box):** the estimated belief–affect–behavior network, where each colored circle is a `NodeInstance` carrying its strength and degree, and each line is an `EdgeInstance` carrying the estimated partial-correlation weight and bootstrap stability summaries. Edge thickness is proportional to $|w|$ (absolute edge weight). Blue = belief (focal), red = affect, green = behavior. **Bottom:** three global `Variable` nodes representing stable variable identities; each `NodeInstance` links to its corresponding `Variable` via `INSTANCE_OF`, allowing the same variable to carry different metric values across contexts.

5.1 Context and network instances

Each estimated network corresponds to a `NetworkInstance` node. A `NetworkInstance` represents one network estimated under a specific estimation context. In the current thesis, the estimation context is defined by survey wave and trust stratum (low vs. high trust in the RKI). These contextual descriptors are represented through `Context` nodes, each storing the wave, trust stratum, and complete-case sample size.

Each `NetworkInstance` is linked to exactly one `Context` node via a `HAS_CONTEXT` relation. Estimation metadata — including the EBIC regularization parameter (γ; see Section 4.1.2) and the correlation input method — are stored as properties on the `NetworkInstance` node itself rather than as separate graph entities. This keeps the schema compact: with six estimation contexts (three waves $\times$ two trust strata), the BKG contains six `NetworkInstance` nodes, each uniquely identified by its context. Robustness specifications ($\gamma = 0.75$, thresholded, Spearman) are evaluated as diagnostic checks (Section 6.8) but are not stored as separate instances in the BKG, since they serve to validate the main specification rather than to represent independent estimation contexts.

5.2 Domain structure of the network

Within each `NetworkInstance`, the CAN-inspired evaluative configuration is represented explicitly. The node set consists of variables belonging to three conceptual domains:

- **Belief domain:** the focal belief perceived `SEVERITY` of COVID-19,
- **Affect domain:** `AFF_FEAR`, `AFF_WORRY`, `WORRY_HEALTH_SYSTEM`, and
- **Behavior domain:** `USE2_MASK`, `USE2_SPACE150`, `USE2_HANDWASH20`, `USE2_AVOID`.

These domains correspond to the belief–affect–behavior configuration described in the theoretical framework. In the graph representation, the variables belonging to these domains appear as nodes within the `NetworkInstance` and are connected through estimated conditional associations.

5.3 Variable identities and node instances

The graph distinguishes between global variable identities and context-specific instantiations of those variables.

- `Variable` nodes represent the stable identity of each survey variable across the entire graph.

- `NodeInstance` nodes represent that variable within a specific `NetworkInstance`.

This distinction is necessary because node-level quantities such as degree or strength depend on the estimated network. A variable such as `SEVERITY` therefore appears once as a global `Variable` node but may appear multiple times through different `NodeInstance` nodes, one for each estimated network instance. Each `NodeInstance` is linked upward to its parent `NetworkInstance` and downward to its corresponding global `Variable` node via an `INSTANCE_OF` relation.

5.4 Edge instances

Conditional associations between variables are represented by `EdgeInstance` nodes. Each `EdgeInstance` belongs to one `NetworkInstance` and connects two `NodeInstance` nodes via typed `FROM` and `TO` relations. The `EdgeInstance` stores the edge-level outputs produced by the estimation pipeline (formally defined in Chapter 4), including

- estimated edge weight and absolute edge weight (point estimate of association strength),

- bootstrap mean and standard deviation (average estimate and variability across resamples),

- percentile interval bounds (2.5th and 97.5th percentiles of the bootstrap distribution; range of observed estimates),

- proportion of non-zero bootstrap estimates (`prop_nonzero`; stability of edge inclusion under regularization), and

- number of bootstrap resamples used ($B = 1000$; reference for interpreting stability summaries).

These quantities are stored to capture not only the estimated strength of an edge, but also its stability under resampling. In regularized network models, individual edge estimates can be sensitive to sampling variability and model selection. The bootstrap summaries therefore provide a practical way to assess how consistently an edge appears and how much its weight varies across resamples.

Including these quantities at the EdgeInstance level allows stability information to be queried and compared across network instances, rather than being treated as a separate post hoc analysis. This design represents estimated relations together with their associated uncertainty and keeps them tied to the specific estimation context.

5.5 Purpose of the representation

The resulting BKG preserves both the internal structure of the estimated belief network and the estimation context under which the network was obtained. A Context identifies a NetworkInstance, which contains the belief, affect, and behavior NodeInstance nodes and their connecting EdgeInstance objects.

The practical advantage of this representation over flat CSV files or relational databases is that it supports structured traversal and comparison across contexts. While CSV files can store the same numerical values, they do not natively represent typed relationships between entities, do not support multi-hop traversal queries, and require custom join logic to link edge properties to their estimation provenance. In contrast, the property-graph structure allows retrieving all edges belonging to a given context in a single traversal, comparing focal-strength values across contexts via pattern-matching queries, and filtering results by estimation metadata (e. g. γ specification or trust stratum) without reconstructing context from column values. These advantages are modest at the scale of six network instances but become more pronounced as the number of contexts, node sets, or estimation specifications grows.

To illustrate, three example Cypher queries are provided below. These queries demonstrate the retrieval operations enabled by the BKG structure and have been formulated to match the exported schema. For each query, an example output computed on the actual BKG is shown to illustrate the retrieval result.

Query 1 — Retrieve the strongest neighbors of SEVERITY in a given network instance, sorted by absolute edge weight:

```
MATCH (net:NetworkInstance {network_id: 'net_g05_w55_low'})
  -[:HAS_NODE_INSTANCE]->(ni:NodeInstance)
  -[:INSTANCE_OF]->(:Variable {node_id: 'SEVERITY'})
WITH net, ni
MATCH (ni)<-[:FROM|TO]-(ei:EdgeInstance)
  -[:FROM|TO]->(ni2:NodeInstance)
  -[:INSTANCE_OF]->(v2:Variable)
WHERE v2.node_id <> 'SEVERITY'
```

```
RETURN v2.node_id AS neighbor,
       ei.abs_weight AS abs_weight,
       ei.prop_nonzero AS prop_nonzero
ORDER BY abs_weight DESC
```

Result (network instance `net_g05_w55_low`):

neighbor	abs_weight	prop_nonzero
WORRY_HEALTH_SYSTEM	0.220	0.86
AFF_FEAR	0.179	0.79
USE2_MASK	0.092	0.55
AFF_WORRY	0.057	0.41
USE2_HANDWASH20	0.039	0.31

Query 2 — Compare focal strength of `SEVERITY` across all low-trust network instances at the main specification:

```
MATCH (ctx:Context {trust_stratum: 'low'})
  <-[:HAS_CONTEXT]-(net:NetworkInstance),
  (net)-[:HAS_NODE_INSTANCE]->(ni:NodeInstance {is_focal: true})
RETURN ctx.wave AS wave,
       ni.strength AS S_SEVERITY
ORDER BY wave
```

Result (low-trust contexts, main specification):

wave	S_SEVERITY
55	0.587
56	0.717
57	0.554

Query 3 — Find the network instance with the highest global strength at the main specification:

```
MATCH (net:NetworkInstance)
RETURN net.network_id AS network_id,
       net.global_strength AS global_strength
ORDER BY global_strength DESC
LIMIT 1
```

Result:

network_id	global_strength
net_g05_w57_low	3.265

These queries operate on stored estimation outputs and metadata. For implementation, the node and relationship tables generated from the estimation outputs were exported as CSV files and imported into a Neo4j-compatible property graph (Neo4j, Inc., 2024). The retrieved values match the corresponding entries in Tables 6.1 and 6.2 reported in Chapter 6, confirming that the BKG stores estimation outputs without transformation.

5.6 Complete workflow overview

Figure 5.2 summarizes the complete workflow from raw COSMO data to conditional mean shift analysis. Each stage produces outputs that feed directly into the next; the BKG layer stores outputs from the estimation stage and serves as input to the query and analysis stages.

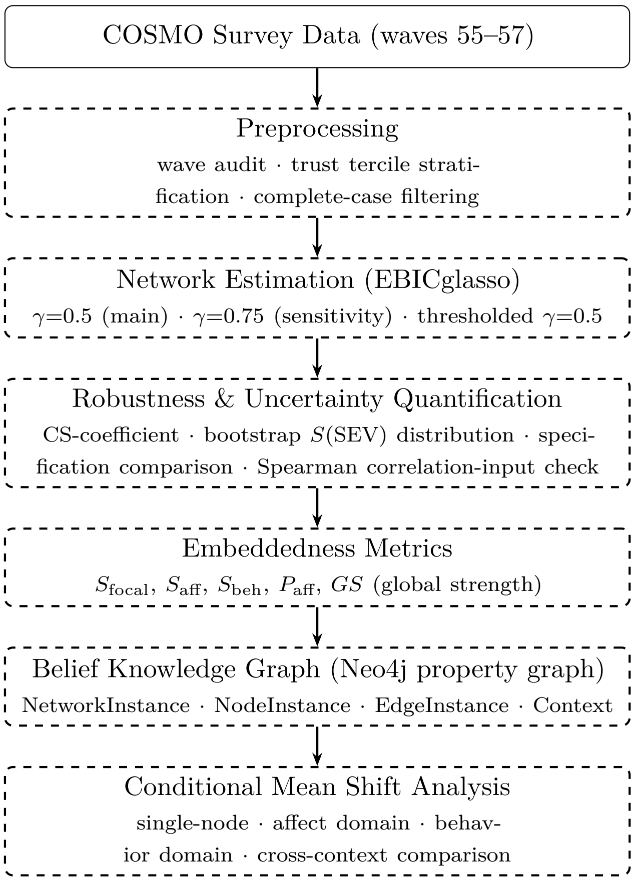

Figure 5.2: Complete workflow from raw COSMO survey data to conditional mean shift analysis.

6 Results

This chapter reports the descriptive properties of the estimated belief networks and their comparison across contexts. As outlined in the Methods chapter, networks were estimated separately across survey waves and trust-defined strata. All reported quantities summarize conditional associations within the estimated Gaussian graphical models and are interpreted descriptively rather than causally.

6.1 Context overview and sample sizes

Networks were estimated for three survey waves (55–57) and two trust strata defined by reported trust in the Robert Koch Institute (RKI): low trust and high trust. This yields six estimation contexts (three waves $\times$ two trust strata), each producing one primary network instance under the main EBICglasso specification ($\gamma = 0.5$, Pearson correlations). These six instances are stored in the BKG.

Three additional robustness specifications were computed for each context: a sensitivity specification with stronger regularization ($\gamma = 0.75$), a thresholded variant ($\gamma = 0.5$ with post-selection significance thresholding), and a Spearman correlation-input re-estimation. These serve as diagnostic checks (Section 6.8) and are not stored in the BKG.

Complete-case sample sizes per context ranged from $n = 309$ (wave 55, high-trust) to $n = 407$ (wave 57, high-trust). Low-trust subsamples were consistently larger than high-trust subsamples within waves 55 and 56 ($n_{\text{high}} = 309, 351$; $n_{\text{low}} = 376, 383$), while in wave 57 the high-trust subsample was larger ($n_{\text{high}} = 407$; $n_{\text{low}} = 336$). These sample sizes represent the respondents remaining after applying the within-wave tercile stratification and complete-case filtering on the node set, as described in Chapter 3. With eight nodes in the fixed node set, the smallest context ($n = 309$) provides a respondent-to-node ratio of approximately 39:1, which is generally considered adequate for regularized GGM estimation (Epskamp et al., 2018).

Table 6.1 summarizes the key structural properties of each network instance under the main specification, providing a reference for the detailed comparisons that follow.

Table 6.1: Context overview: structural properties of the six main-specification network instances ($\gamma = 0.5$). Bridge edges are defined here as edges connecting a node from the belief or affect domain to a node from the behavior domain.

Wave	Trust	n	Edges	Bridge edges	GS	S(SEV)
55	low	376	23	11	3.177	0.587
55	high	309	25	13	2.948	0.753
56	low	383	21	9	3.049	0.717
56	high	351	21	10	2.656	0.503
57	low	336	22	10	3.265	0.554
57	high	407	17	6	2.303	0.364

6.2 Structural properties of the wave 57 high-trust network

The wave 57 high-trust network warrants specific attention before presenting cross-context comparisons. Under the main specification ($\gamma = 0.5$), this context retained 17 nonzero edges with a global strength of $GS = 2.303$ — the lowest among the six main-specification contexts but not structurally anomalous. Six bridge edges survived regularization, and `SEVERITY` retained three conditional associations: with `AFF_FEAR` ($|\hat{w}| = 0.229$), `AFF_WORRY` ($|\hat{w}| = 0.094$), and `USE2_HANDWASH20` ($|\hat{w}| = 0.041$). The $\gamma = 0.75$ sensitivity specification produced an identical network for this context (same 17 edges, same weights).

The picture changes under the thresholded specification ($\gamma = 0.5$ with post-selection Fisher z-test thresholding): only 6 edges survive, reducing to a single bridge edge, and `SEVERITY` retains only its association with `AFF_FEAR`. This contrasts with 9–14 edges in other contexts under the thresholded specification. The thresholded wave 57 high-trust result therefore indicates that many of the main-specification edges in this context have weaker statistical support and should be treated with additional caution. Descriptive statistics for behavioral items (Appendix D) show higher means and lower standard deviations in the high-trust stratum, consistent with ceiling effects that may attenuate Pearson correlations in this context. These two explanations — genuine lower coupling and ceiling-effect attenuation — cannot be distinguished in the present design.

Cross-context comparisons involving the wave 57 high-trust network should therefore be understood as comparing a context with weaker and fewer surviving associations to the other contexts, without attributing this pattern unambiguously to either substantive or methodological causes.

6.3 Focal neighborhood of perceived COVID-19 severity

The primary analytical focus of the networks is the node representing perceived COVID-19 severity (`SEVERITY`). To examine how this focal belief is embedded within the broader evaluative system, the edges incident to the `SEVERITY` node were extracted and compared across contexts.

Figure 6.1 visualizes the estimated edge weights connecting `SEVERITY` to the affective and behavioral variables included in the model. Rows correspond to neighboring variables, whereas columns represent estimation contexts defined by wave, trust stratum, and specification.

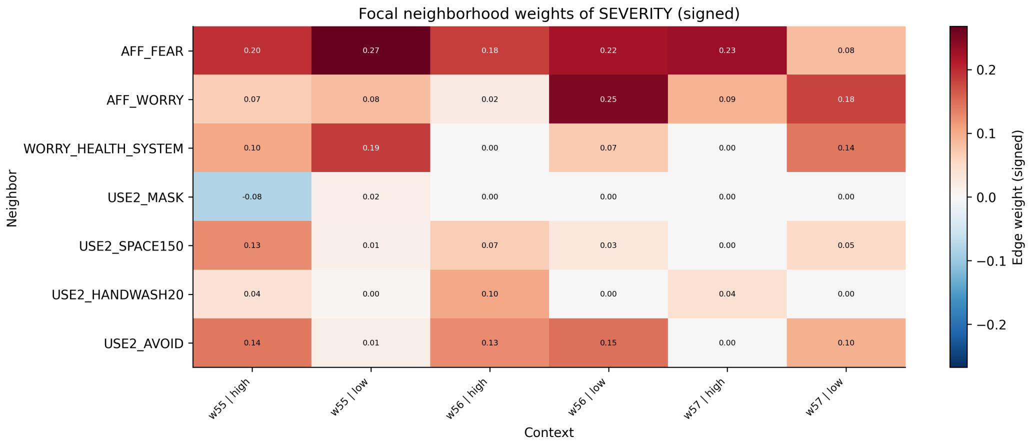

Figure 6.1: Focal neighborhood of `SEVERITY` across contexts. Colors represent the estimated conditional edge weights between `SEVERITY` and neighboring variables in each network. Positive values indicate positive partial correlations after controlling for all other nodes in the set.

Across all estimated networks, the strongest edge incident to `SEVERITY` always involved an affective variable rather than a behavioral variable. More specifically, the strongest neighbor of `SEVERITY` was always either `AFF_FEAR` or `AFF_WORRY`, indicating that the focal belief was most strongly linked to affective responses in every estimated context. Behavioral variables such as mask wearing or avoidance behaviors were also connected to `SEVERITY`, but their edge weights were generally smaller in magnitude.

Three structural patterns were consistent across the estimated networks:

Finding 1: Affect coupling dominates behavior coupling. Across all contexts, the proportion of `SEVERITY`'s total strength attributable to affect connections (P_{aff}) exceeded the proportion attributable to behavior connections (P_{beh}) in four of six contexts. Under the main specification ($\gamma = 0.5$), SEVERITY's affect share ranged from approximately 41% to 91% across contexts. This pattern indicates that perceived severity was more tightly conditionally associated with fear and worry responses than with protective behavioral tendencies, after controlling for all other shared variance. Two high-trust contexts qualify as exceptions: wave 56 high-trust ($P_{\text{aff}} = 0.411$) and wave 55 high-trust ($P_{\text{aff}} = 0.485$), where the behavior component was slightly larger than the affect component. In the wave 55 high-trust case the split is close to even, while wave 56 high-trust shows a clearer behavior dominance. Within the CAN framework, the overall pattern is consistent with affect operating as an intermediate element between severity beliefs and behavior.

Finding 2: USE2_AVOID is SEVERITY's most frequent dominant behavioral neighbor. Among all six contexts, avoidance behavior (USE2_AVOID) was SEVERITY's strongest behavioral neighbor in four contexts: wave 55 high-trust, wave 56 low-trust, wave 56 high-trust, and wave 57 low-trust. In wave 55 low-trust, the dominant behavioral neighbor was USE2_MASK, though all four behavioral edges in that context were small in magnitude and classified as moderate or unstable by the bootstrap stability criterion, indicating limited reliability. In wave 57 high-trust, the single surviving behavioral edge connected SEVERITY to USE2_HANDWASH20 ($|\hat{w}| = 0.041$). Thus, while avoidance behavior was the most frequent dominant behavioral neighbor of the focal belief across contexts, its dominance was clearest in waves 56 and 57. A possible — though speculative — interpretation is that avoidance requires a more direct and ongoing risk appraisal than habitual hygiene behaviors, which may become less contingent on severity beliefs over the course of a prolonged pandemic; this cannot be tested within the present design.

Finding 3: Low-trust networks show consistently higher global strength. Across all three waves, global strength was higher in the low-trust stratum than in the high-trust stratum (GS: 3.177 vs. 2.948 at wave 55; 3.049 vs. 2.656 at wave 56; 3.265 vs. 2.303 at wave 57). Focal strength of SEVERITY was higher in the low-trust stratum in waves 56 and 57, but in wave 55 the pattern was reversed ($S = 0.753$ for high-trust vs. $S = 0.587$ for low-trust), driven by stronger high-trust connections to avoidance behavior and to WORRY_HEALTH_SYSTEM in that wave. These results indicate that global connectivity was consistently higher in low-trust contexts, while focal coupling showed a less uniform pattern that varied by wave. One possible account is that respondents reporting lower institutional trust lack an external referent for threat appraisal and therefore rely more heavily on the internal coherence of their own belief–affect–behavior system, producing tighter coupling among its components; an alternative is that the pattern reflects response-style differences rather than genuine differences in psychological organization. The present design cannot distinguish between these accounts. A fuller discussion is provided in Section 8.2.

Although the identity of the strongest neighbor varied between `AFF_FEAR` and `AFF_WORRY` across contexts, the dominance of affective variables among the largest focal edges remained stable across waves and trust strata. This structural stability suggests that the local neighborhood of the focal belief was characterized primarily by affective coupling rather than direct behavioral associations. Figures 6.2 and 6.3 illustrate the estimated network structure for waves 55 and 57 respectively, showing the low- and high-trust network side by side for each wave.

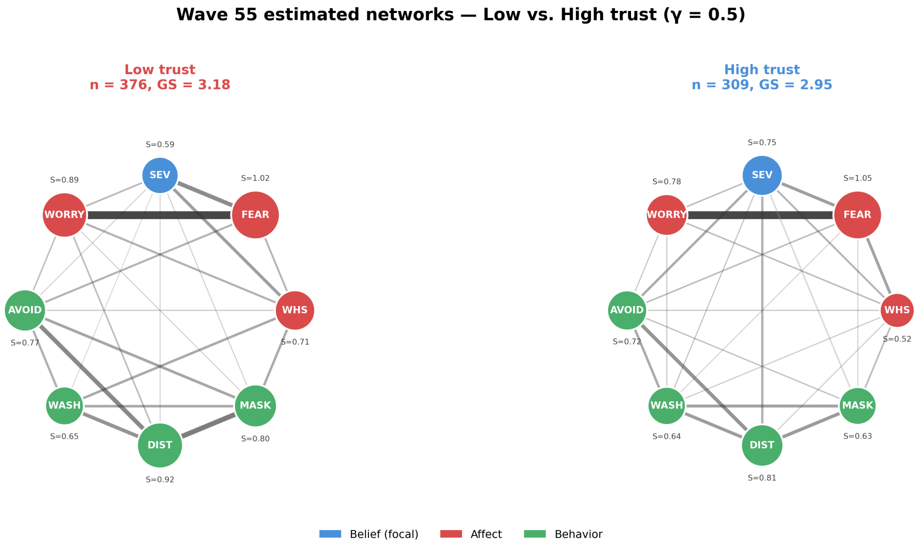

Figure 6.2: Wave 55 estimated networks for low-trust (left, $n = 376$, $GS = 3.18$) and high-trust (right, $n = 309$, $GS = 2.95$) subgroups under the main specification ($\gamma = 0.5$). Node size is proportional to node strength; edge thickness is proportional to absolute partial-correlation weight. Node color indicates domain: blue = belief (focal), red = affect, green = behavior. The low-trust network is denser at the global level; focal strength is higher in the high-trust network in this wave.

In the following sections, this local structure is summarized using embeddedness metrics that quantify the overall strength and composition of `SEVERITY`'s connections within each network.

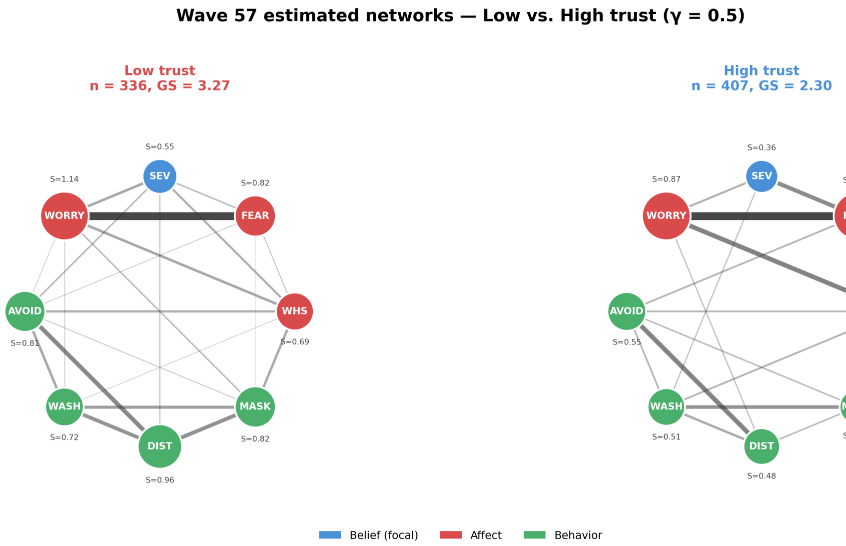

Figure 6.3: Wave 57 estimated networks for low-trust (left, $n = 336$, $GS = 3.27$) and high-trust (right, $n = 407$, $GS = 2.30$); visual encoding as in Figure 6.2. The high-trust network shows lower overall connectivity, retaining 17 edges with 6 bridge connections; the `SEVERITY` node retains three edges (to `AFF_FEAR`, `AFF_WORRY`, and `USE2_HANDWASH20`). Under the thresholded specification, only 6 edges survive for this context, indicating that many main-specification edges have limited statistical support.

6.4 Embeddedness metrics of the focal belief

To quantify the position of perceived COVID-19 severity within the evaluative system, the connectivity-based metrics defined in Section 4.3 were computed for each network instance. Focal strength S(SEVERITY) (Equation 4.1) captures the overall magnitude of conditional associations linking the focal belief to surrounding affect and behavior nodes: larger values indicate stronger coupling. Focal strength was further decomposed into the affect-allocated component S_{aff} (Equation 4.2) and behavior-allocated component S_{beh} (Equation 4.3), and the affect share P_{aff} (Equation 4.4) summarizes the relative dominance of affective vs. behavioral coupling. Figure 6.4 visualizes P_{aff} across contexts.

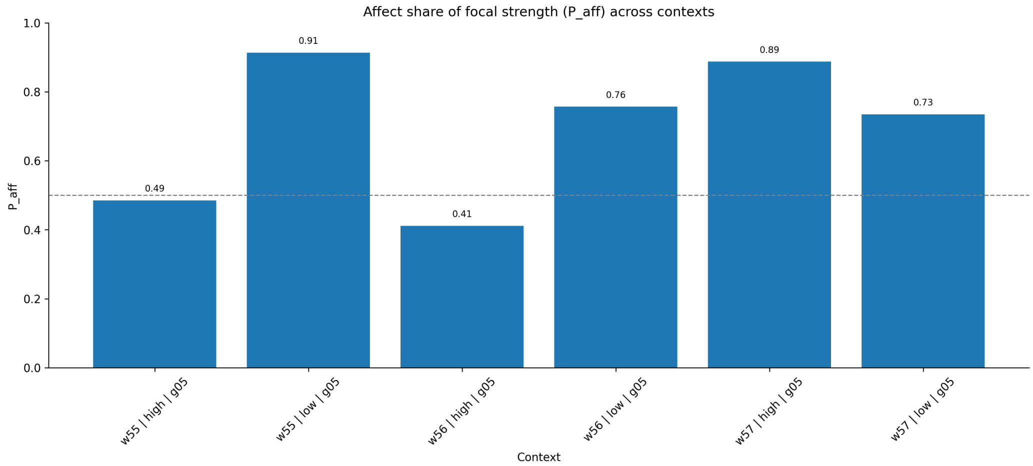

Figure 6.4: Affective share of focal coupling (P_{aff}) across estimation contexts. Values above 0.5 indicate that affective variables contribute more strongly than behavioral variables to the conditional coupling of `SEVERITY`.

Across contexts, affective variables accounted for a substantial portion of the focal belief's coupling. In several contexts, the affective share exceeded 0.7, indicating marked affective dominance in the local neighborhood of `SEVERITY`.

At the same time, the composition of focal coupling was not identical across contexts. Low-trust networks tended to show a more strongly affect-dominated structure, whereas some high-trust networks showed a more balanced distribution between affective and behavioral contributions.

These results indicate that perceived COVID-19 severity was consistently embedded in a joint affective-behavioral network, but that the local composition of this embeddedness varied across estimation contexts.

To examine the internal composition of focal coupling at a finer resolution than P_{aff}, Figure 6.5 decomposes S(SEVERITY) into the individual edge-weight contributions of each neighbor.

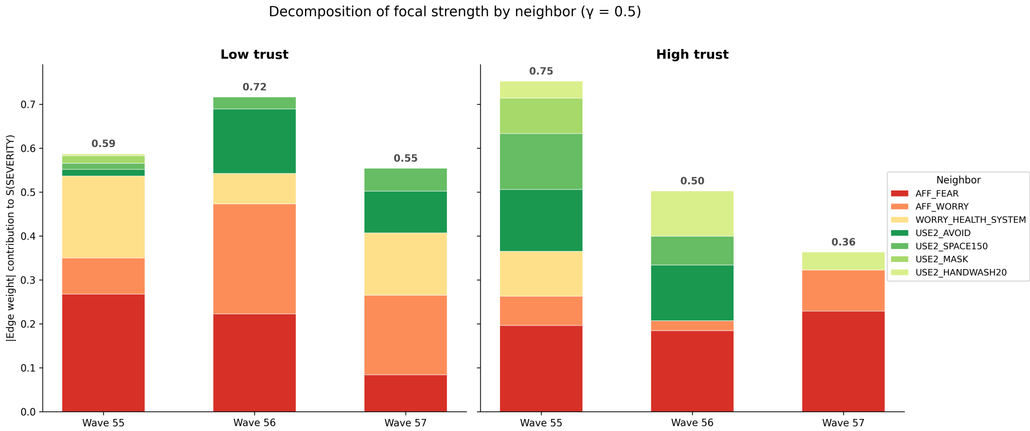

Figure 6.5: Decomposition of focal strength S(SEVERITY) by individual neighbor, shown separately for low-trust (left) and high-trust (right) contexts under the main specification ($\gamma = 0.5$). Warm colors (red/orange/yellow) represent affective neighbors; green shades represent behavioral neighbors. The bold number above each bar is the total focal strength. Low-trust contexts show a broadly stable total (≈ 0.55–0.72) with shifting internal composition across waves. High-trust contexts show a progressive decline from 0.75 (wave 55) to 0.36 (wave 57), with behavioral contributions diminishing substantially by wave 57.

Several patterns emerge from the decomposition. First, `AFF_FEAR` (red) provides a stable contribution across most contexts. Second, the identity of the dominant affective neighbor shifts: `WORRY_HEALTH_SYSTEM` contributes substantially in wave 55 low-trust but recedes in later waves, while `AFF_WORRY` becomes the largest contributor in wave 56 and 57 low-trust networks. Third, the behavioral segments (green shades) shift toward `USE2_AVOID` from wave 56 onward. In wave 57 high-trust, `AFF_FEAR` is the dominant neighbor, with `AFF_WORRY` and `USE2_HANDWASH20` providing smaller residual contributions.

6.5 Global network connectivity across contexts

Beyond the local structure surrounding the focal belief, the overall connectivity of the estimated networks was summarized using global strength GS (Equation 4.5), which provides a descriptive index of the overall magnitude of conditional associations among the variables in the model. Figure 6.6 shows GS values across survey waves and trust strata.

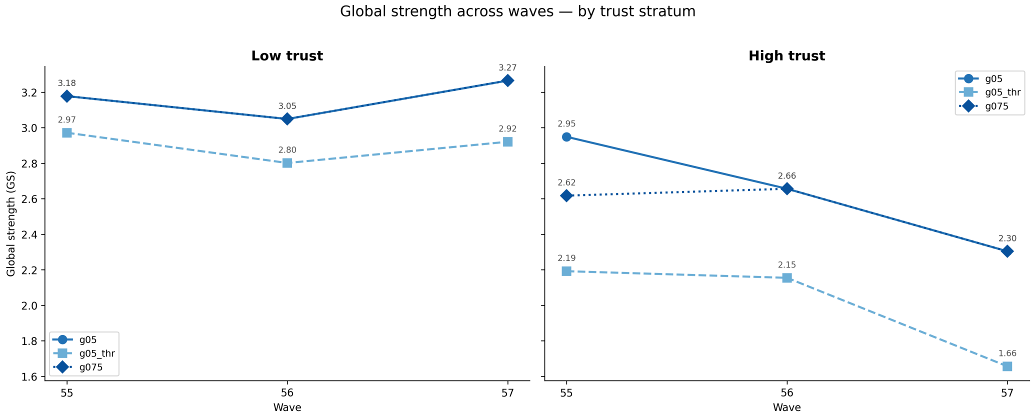

Figure 6.6: Global network strength (GS) across waves, shown separately for low-trust (left) and high-trust (right) contexts. Low-trust GS remains stable across waves (≈ 3.0–3.3), whereas high-trust GS declines from wave 55 to wave 57.

Across all waves, networks estimated within the low-trust stratum exhibited higher global strength than those estimated within the high-trust stratum. This indicates that the belief–affect–behavior variables included in the model showed higher summed partial-correlation magnitudes among respondents reporting lower trust in the Robert Koch Institute.

In addition to this cross-stratum difference, the high-trust networks displayed a decline in global strength across waves, from $GS = 2.948$ at wave 55 to $GS = 2.303$ at wave 57. In contrast, the low-trust networks remained comparatively stable across waves ($GS = 3.18$, 3.05, 3.27 for waves 55–57 respectively).

These differences suggest that the evaluative system represented by the included variables showed higher summed partial-correlation magnitudes among low-trust respondents than among high-trust respondents. Because the data are repeated cross-sectional, wave-to-wave differences in global strength describe changes in estimated conditional association structure between distinct respondent samples and do not imply longitudinal change within individuals.

To provide a system-wide view of how node-level connectivity varies across contexts, Figure 6.7 displays the strength of every variable in each estimation context under the main specification.

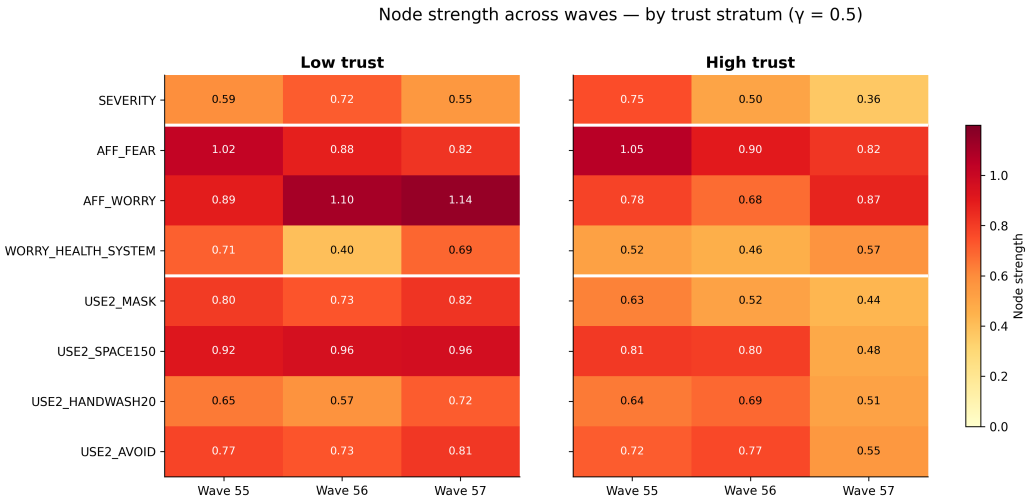

Figure 6.7: Node strength (sum of absolute incident edge weights) for all eight variables across estimation contexts under the main specification ($\gamma = 0.5$). Low-trust contexts (left panel) show uniformly warm colors across all three waves, indicating stable coupling throughout the system. High-trust contexts (right panel) show a gradual decline from wave 55 to wave 57, with the wave 57 column showing reduced but non-zero coupling across all nodes. White horizontal lines separate the belief, affect, and behavior domains.

The heatmap shows that the wave 57 high-trust reduction in connectivity is not specific to the focal belief: most nodes show reduced strength relative to earlier waves, indicating a system-wide weakening of coupling rather than a selective disconnection of `SEVERITY`.

Table 6.2 summarizes the key embeddedness metrics per network instance under the main specification.

Table 6.2: Embeddedness metrics per network instance (main specification, $\gamma = 0.5$). S: focal strength of `SEVERITY`; P_{aff}: affect share; GS: global strength; n_{edges}: number of nonzero edges.

Wave	Trust	n	n_{edges}	S(SEV)	P_{aff}	GS
55	low	376	23	0.587	0.914	3.177
55	high	309	25	0.753	0.485	2.948
56	low	383	21	0.717	0.757	3.049
56	high	351	21	0.503	0.411	2.656
57	low	336	22	0.554	0.735	3.265
57	high	407	17	0.364	0.888	2.303

6.6 Trust-stratum comparison of focal embeddedness

To examine whether the embeddedness of the focal belief differed between trust contexts, focal coupling metrics were compared within each survey wave across the low- and high-trust strata.

Global strength was higher in the low-trust stratum in every wave. Focal strength of `SEVERITY` was higher in the low-trust stratum in waves 56 and 57 ($S = 0.717$ vs. 0.503 and $S = 0.554$ vs. 0.364, respectively). In wave 55, the pattern was reversed: focal strength was higher in the high-trust stratum ($S = 0.753$ vs. 0.587), driven by stronger conditional associations between `SEVERITY` and behavioral variables in that context. Thus, the trust-stratum contrast in focal coupling was consistent in two of three waves and reversed in one, while the contrast in global coupling was consistent across all waves.

6.7 Belief knowledge graph: stored network instances

As described in Chapter 5, the estimated network instances were encoded in a belief knowledge graph (BKG) to enable structured storage, retrieval, and comparison. This section illustrates the resulting representation using an example network instance and demonstrates the query operations that the BKG enables.

6.7.1 Example: a single network instance in the BKG

Figure 6.8 shows the wave 55 low-trust network instance ($\gamma = 0.5$) rendered with its estimated edge weights and node-strength values. This is the same statistical network as the left panel of Figure 6.2, but annotated with the quantities that are stored as properties in the BKG: each `NodeInstance` carries its `strength` and `degree`, and each `EdgeInstance` carries the estimated `weight`, `abs_weight`, and bootstrap stability summaries.

In the BKG, this network instance is stored as a `NetworkInstance` node linked to its `Context` (wave 55, low trust, $n = 376$), with estimation metadata ($\gamma = 0.5$, Pearson correlation input) stored as properties on the `NetworkInstance`. The instance contains eight `NodeInstance` nodes (one per variable) and 23 `EdgeInstance` nodes (one per nonzero edge). Each `NodeInstance` links upward to its parent `NetworkInstance` via `HAS_NODE_INSTANCE` and downward to its global `Variable` identity via `INSTANCE_OF`. This structure ensures that the same variable (e. g. `SEVERITY`) can carry different strength values in different contexts without ambiguity.

6.7.2 Cross-context retrieval via Cypher queries

The practical value of the BKG representation lies in its support for structured retrieval across contexts. Figure 6.9 illustrates an example query that retrieves the focal strength of `SEVERITY` across all six main-specification contexts in a single traversal.

The values returned by this query correspond exactly to the $S($`SEVERITY`$)$ estimates reported in Table 6.2, confirming that the BKG stores estimation outputs without transformation and supports declarative retrieval with provenance preserved.

This query demonstrates three properties of the BKG that go beyond flat-file storage. First, the traversal is *context-aware*: the result explicitly links each strength value to its estimation context (wave and trust stratum) and specification (γ). Second, the query is *declarative*: the same pattern can be parameterized to retrieve any node metric for any variable across any subset of contexts, without custom join logic. Third, the result is *provenance-preserving*: the returned values are properties of specific `NodeInstance` objects, each traceable to one `NetworkInstance` with known estimation settings, sample size, and bootstrap stability summaries. Additional query

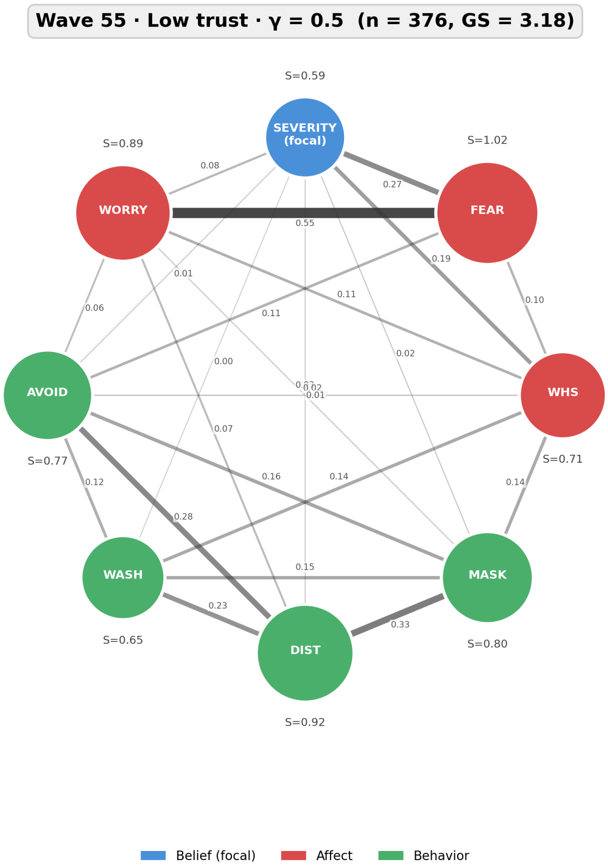

Figure 6.8: A single network instance (wave 55, low trust, $\gamma = 0.5$) rendered with stored BKG properties. Node labels show node strength (S); edge labels show estimated partial-correlation weights. Blue = focal belief; red = affect; green = behavior. This instance has 23 nonzero edges, $GS = 3.18$, and a focal strength of $S(\text{SEVERITY}) = 0.587$.

BKG query example: retrieving SEVERITY focal strength across contexts

Cypher Query

```
MATCH (ni:NodeInstance)-[:INSTANCE_OF]->(v:Variable {node_id: 'SEVERITY'}),
(net:NetworkInstance)-[:HAS_NODE_INSTANCE]->(ni),
(net)-[:HAS_CONTEXT]->(ctx:Context)
RETURN ctx.wave AS wave, ctx.trust_stratum AS trust,
ni.strength AS S_SEVERITY
ORDER BY wave, trust;
```

↓ returns

wave	trust	S_SEVERITY
55	high	0.753
55	low	0.587
56	high	0.503
56	low	0.717
57	**high**	**0.364**
57	low	0.554

Highlighted: wave 57 high-trust — lowest focal strength (S = 0.364)

Figure 6.9: Example Cypher query and its result. The query traverses from `NodeInstance` through `INSTANCE_OF` to the global `Variable` node for `SEVERITY`, then links through `NetworkInstance` to its `Context`. The result table returns one row per context with the context-specific focal strength. The highlighted row (wave 57, high trust, $S = 0.364$) corresponds to the context with the lowest focal strength, discussed in Section 6.2.

examples are provided in Appendix C. Beyond retrieval, the correlation matrices stored alongside each network instance also serve as the input for the conditional mean shift analysis in Chapter 7, where the estimated joint distribution is used to compute implied displacements of the focal belief under domain-level conditioning.

6.8 Robustness and stability assessment

This section consolidates all robustness checks and stability diagnostics for the estimated networks. Four dimensions of robustness are assessed: (i) regularization strength ($\gamma = 0.75$ vs. $\gamma = 0.5$), (ii) post-selection significance thresholding, (iii) correlation input (Spearman vs. Pearson), and (iv) resampling-based stability (edge-level bootstrap, focal-strength bootstrap, and case-dropping stability). The methods underlying each check are defined in Chapter 4; the present section reports the empirical results.

6.8.1 Regularization strength: $\gamma = 0.75$ sensitivity specification

Under the $\gamma = 0.75$ sensitivity specification, EBICglasso selected identical networks in five of six contexts relative to the $\gamma = 0.5$ main specification — the same edges with the same weights. The exception was wave 55 high-trust, where $\gamma = 0.75$ produced a sparser network ($S(\text{SEV}) = 0.595$ vs. 0.753; $GS = 2.617$ vs. 2.948), indicating that some edges in that context were sensitive to the penalization level. The Spearman rank correlation between $\gamma = 0.5$ and $\gamma = 0.75$ focal strength orderings across the six contexts is $r_s = 0.943$ ($p = .005$), confirming near-perfect ordinal agreement.

6.8.2 Post-selection thresholding

Under the thresholded specification ($\gamma = 0.5$ with post-selection Fisher z-test thresholding at $p > 0.05$), smaller edge weights were set to zero, reducing the density of some networks. Despite this, the substantive patterns remained qualitatively stable: affective variables continued to dominate the focal neighborhood of `SEVERITY` in all contexts, low-trust contexts retained higher global connectivity than high-trust contexts, and the relative ordering of network instances by GS magnitude was preserved.

Table 6.3 provides a complete metric comparison across all three γ-based specifications for each context.

Table 6.3: Embeddedness metric comparison across estimation specifications for all six contexts. Thr = thresholded $\gamma = 0.5$. Spearman rank correlation between $\gamma = 0.5$ and $\gamma = 0.75$ focal strength orderings: $r_s = 0.943$, $p = .005$.

		S(SEV)			GS		
Wave	Trust	γ=0.5	γ=0.75	Thr	γ=0.5	γ=0.75	Thr
55	low	0.587	0.587	0.464	3.177	3.177	2.971
55	high	0.753	0.595	0.526	2.948	2.617	2.192
56	low	0.717	0.717	0.632	3.049	3.049	2.801
56	high	0.503	0.503	0.352	2.656	2.656	2.154
57	low	0.554	0.554	0.326	3.265	3.265	2.920
57	high	0.364	0.364	0.245	2.303	2.303	1.656

These results indicate that the main descriptive findings were not driven by a single edge-selection setting, but remained qualitatively stable under the more conservative thresholded estimation.

6.8.3 Correlation input sensitivity: Spearman check

To assess whether the structural findings depend on the Pearson approximation for ordinal data (see Section 4.1.1), the estimation pipeline was re-run for all six main-specification contexts using Spearman rank correlations as input instead of Pearson correlations. The key comparison criterion is whether the composition and ranking of `SEVERITY`'s focal neighborhood — specifically, whether affective variables rank above behavioral variables — is preserved under Spearman input.

Table 6.4 reports the focal strength, global strength, and the identity of the top-ranked neighbor of `SEVERITY` under both correlation inputs for each context.

The top-ranked neighbor of `SEVERITY` was identical under Pearson and Spearman inputs in five of six contexts. The single divergence occurred in wave 56 low-trust, where the Pearson specification ranked `AFF_WORRY` highest while the Spearman specification ranked `AFF_FEAR` highest; both are affective variables, so the qualitative pattern of affective dominance was preserved even in this context. The Spearman rank correlation between Pearson and Spearman focal strength orderings across the six contexts was $r_s = 0.886$ ($p = .019$), indicating strong ordinal agreement. Global strength values under Spearman input were closely aligned with Pearson values in most contexts, with the largest divergence in wave 57 high-trust (GS: 2.303 vs. 1.999), where the Spearman specification produced a sparser network. These results indicate that the main structural findings — in particular, the affective dominance of

Table 6.4: Correlation-input sensitivity: Pearson vs. Spearman re-estimation of the main specification ($\gamma = 0.5$) for all six contexts. Top neighbor reports the variable with the largest absolute edge weight to `SEVERITY`. The Spearman rank correlation between Pearson and Spearman focal strength orderings is $r_s = 0.886$ ($p = .019$).

		S(SEV)		GS		Top neighbor
Wave	Trust	Pearson	Spearman	Pearson	Spearman	(Pearson / Spearman)
55	low	0.587	0.571	3.177	3.333	AFF_FEAR / AFF_FEAR
55	high	0.753	0.591	2.948	2.687	AFF_FEAR / AFF_FEAR
56	low	0.717	0.683	3.049	3.040	AFF_WORRY / AFF_FEAR
56	high	0.503	0.531	2.656	2.810	AFF_FEAR / AFF_FEAR
57	low	0.554	0.506	3.265	3.131	AFF_WORRY / AFF_WORRY
57	high	0.364	0.315	2.303	1.999	AFF_FEAR / AFF_FEAR

the focal neighborhood and the trust-stratum asymmetry in global strength — are not sensitive to the Pearson-on-ordinal approximation.

6.8.4 Bootstrap uncertainty of focal strength

To assess whether the primary embeddedness measure is robust to sampling variability, the focal strength S(`SEVERITY`) was recomputed across all 1000 bootstrap resamples per context (see Section 4.2.2 for method details). Table 6.5 reports the resulting distributions.

In most contexts, the bootstrap mean tracks the point estimate closely, with moderate variability (SD ranging from 0.077 to 0.139). The wave 55 high-trust context shows the largest discrepancy between the point estimate (0.753) and the bootstrap mean (0.584), indicating that the point estimate may overstate the typical resampled focal strength in that context. The negative 2.5th-percentile bound (-0.082) for wave 55 high-trust is an artifact of the aggregation method described in Section 4.2.2 and does not reflect a plausible focal-strength value.

6.8.5 Case-dropping stability

Table 6.6 reports the CS-coefficient for strength centrality per context (see Section 4.2.3 for method details). Five of the six contexts meet the conventional threshold of 0.50. The exception is wave 55 high-trust (CS = 0.440), which falls below the threshold. Strength centrality rankings in that context should therefore be interpreted with additional caution, as they are more sensitive to the composition

Table 6.5: Bootstrap distribution of focal strength S(SEVERITY) per network context (main specification, $\gamma = 0.5$, $B = 1000$). Point estimate, bootstrap mean, combined bootstrap SD, and summed 2.5th/97.5th percentile bounds are reported. Because the interval bounds were computed by summing edge-level percentiles rather than by taking percentiles of the bootstrap sum distribution, they represent conservative bounding heuristics. In particular, summed lower bounds can fall below zero even though S(SEVERITY) is non-negative by definition (see Section 4.2.2 for discussion). All summaries should be treated as exploratory stability indicators rather than classical confidence intervals.

Wave	Trust	Point est.	Boot. mean	Boot. SD	2.5th pct	97.5th pct
55	low	0.587	0.601	0.110	0.237	1.229
55	high	0.753	0.584	0.139	−0.082	1.244
56	low	0.717	0.694	0.101	0.306	1.148
56	high	0.503	0.515	0.104	0.121	0.988
57	low	0.554	0.530	0.112	0.098	1.022
57	high	0.364	0.349	0.077	0.128	0.625

of the sample. The stability pattern is not monotonically determined by sample size: wave 56 low-trust ($n = 383$) achieves the highest CS (0.671) despite not being the largest sample, and wave 57 low-trust ($n = 336$, smaller than three other contexts) matches the stability of the largest context (wave 57 high-trust, $n = 407$, CS = 0.595). Within the high-trust group, sample size and CS do co-vary directionally — the smallest high-trust context (wave 55, $n = 309$) is the sole below-threshold case — but this pattern does not hold across all six contexts. Taken together, sample size appears to be a contributing factor but correlation structure also plays a role in determining stability differences across contexts.

Table 6.6: Case-dropping stability (CS-coefficient for strength centrality) per network context (main specification, $\gamma = 0.5$, $B = 1000$). Values ≥ 0.50 are conventionally interpreted as acceptable stability.

Wave	Trust	n	CS-coefficient	Interpretation
55	low	376	0.516	Acceptable
55	high	309	0.440	Below threshold
56	low	383	0.671	Acceptable
56	high	351	0.516	Acceptable
57	low	336	0.595	Acceptable
57	high	407	0.595	Acceptable

6.8.6 Robustness summary

Across the four robustness dimensions, the main structural findings proved stable. The $\gamma = 0.75$ specification produced identical networks in five of six contexts (the exception being wave 55 high-trust, where some edges were removed). The thresholded specification reduced network density but preserved the qualitative patterns (affective dominance, trust-stratum asymmetry in GS, relative ordering of focal strength). The Spearman correlation-input check preserved the focal neighborhood composition in five of six contexts ($r_s = 0.886$, $p = .019$; the single divergence involved two affective variables swapping rank in wave 56 low-trust). Bootstrap summaries showed moderate sampling variability in focal strength but no reversals of the key structural patterns. Case-dropping stability was acceptable in five of six contexts, with wave 55 high-trust as the sole exception. The one context requiring interpretive caution across multiple diagnostics is wave 55 high-trust, which showed below-threshold CS-coefficient stability and the largest bootstrap discrepancy between point estimate and resampled mean.

Table 6.7 consolidates the robustness profile of each context across all four diagnostic dimensions for quick cross-context comparison.

Table 6.7: Robustness profile per estimation context across all four diagnostic dimensions. γ-stability: identical network under $\gamma = 0.75$ vs. $\gamma = 0.5$ (Yes/No). Thr-stability: qualitative patterns preserved under thresholded specification (Yes). Spearman: top neighbor of `SEVERITY` preserved under Spearman input. CS: case-dropping coefficient ≥ 0.50 (acceptable) or < 0.50 (below threshold).

Wave	Trust	γ-stability	Thr-stability	Spearman	CS ($\geq$0.50)
55	low	Yes	Yes	Yes	Yes (0.516)
55	high	No	Yes	Yes	No (0.440)
56	low	Yes	Yes	No[a]	Yes (0.671)
56	high	Yes	Yes	Yes	Yes (0.516)
57	low	Yes	Yes	Yes	Yes (0.595)
57	high	Yes	Yes	Yes	Yes (0.595)

[a] Top neighbor shifts from `AFF_WORRY` (Pearson) to `AFF_FEAR` (Spearman); both are affective variables, so the qualitative pattern of affective dominance is preserved.

6.9 Summary of empirical findings

This chapter described the structural properties of the estimated belief networks and their variation across survey waves and trust-defined contexts. Several descriptive patterns emerged.

First, the local neighborhood of the focal belief (perceived COVID-19 severity) was consistently dominated by affective variables. Across all estimated networks, the strongest neighbor of `SEVERITY` was always either `AFF_FEAR` or `AFF_WORRY`, indicating that perceived severity was most strongly linked to affective responses rather than to protective behaviors within the conditional dependence structure.

Second, embeddedness metrics showed that affective variables accounted for a large share of the focal belief's coupling in most contexts. In four of six contexts, $P_{\text{aff}} > 0.5$; the two exceptions were wave 55 high-trust ($P_{\text{aff}} = 0.485$, approximately balanced) and wave 56 high-trust ($P_{\text{aff}} = 0.411$), where behavioral coupling was slightly larger than affective coupling.

Third, systematic differences were observed between trust strata at the level of global strength. Networks estimated for respondents reporting lower trust in the Robert Koch Institute exhibited higher summed partial-correlation magnitudes in all three waves. Focal strength showed a similar pattern in waves 56 and 57, but was higher in the high-trust stratum in wave 55, indicating that the trust-stratum contrast in focal coupling is not uniform across waves.

Fourth, these descriptive patterns were stable across all four robustness dimensions. The specification comparison (Table 6.3) showed $r_s = 0.943$ ($p = .005$) between $\gamma = 0.5$ and $\gamma = 0.75$ focal strength orderings. Metric-level bootstrap uncertainty (Table 6.5) showed moderate sampling variability without reversals. Case-dropping stability (Table 6.6) was acceptable in five of six contexts; wave 55 high-trust was the single exception (CS = 0.440). The Spearman correlation-input check confirmed that the main structural findings are not driven by the Pearson approximation for ordinal data.

In sum, these findings characterize how the focal belief was embedded within the belief–affect–behavior system across contexts. The implications of these patterns for understanding belief structure and belief systems are considered in the following chapters.

7 Conditional Mean Shift Analysis

7.1 Rationale and framing

The embeddedness metrics reported in Chapter 6 characterize the static conditional association structure of the estimated networks. To illustrate a downstream use of this structure, the present chapter introduces a *conditional mean shift analysis*: using the estimated GGM as a probabilistic model of the joint distribution of evaluative components, we examine how the conditional expectation of a node changes when a subset of nodes is fixed to an extreme value.

This is not a causal claim. No intervention is simulated and no directional influence is assumed. The analysis is a statement about the *associational structure* of the estimated Gaussian joint distribution: given the modeled covariance structure, what is the implied conditional mean of `SEVERITY` when the affect domain or behavior domain is set to an extreme value? Comparisons across trust-stratified networks illustrate how this implied displacement differs as a function of the structural context in which the focal belief is embedded.

7.2 Method

7.2.1 Statistical foundation

For a multivariate Gaussian with mean vector μ and covariance matrix Σ, the conditional distribution of a variable subset Y given $X = x$ is:

$$\mu_{Y|X=x} = \mu_Y + \Sigma_{YX}\Sigma_{XX}^{-1}(x - \mu_X). \tag{7.1}$$

The implied conditional displacement of node Y_i is defined as the change in conditional expectation relative to its mean:

$$\Delta_i = \mu_{Y_i|X=x} - \mu_{Y_i}. \tag{7.2}$$

The conditioning value is set to $x = \mu_X + c\,\sigma_X$ with $c = +1$ (one standard deviation above the mean). Because the GGM is estimated on standardized variables (Pearson correlation inputs), $\mu = \mathbf{0}$ and $\Sigma = R$, the within-context Pearson correlation matrix. Consequently, Δ_i is expressed in standard deviation units relative to the within-context distribution.

In concrete terms, the analysis evaluates how the estimated correlation structure translates into expected changes in the focal belief when the conditioning variables are set one standard deviation above their mean. A value of $\Delta_{\texttt{SEVERITY}} = 0.66$ means the model implies that the severity belief would lie approximately two-thirds of a standard deviation above its mean under this condition. Larger values indicate stronger coupling between the conditioning domain and the focal belief. Comparing affect-domain and behavior-domain Δ values reveals which domain is more strongly associated with the severity belief, and comparisons across trust strata indicate whether this association differs between subgroups.

7.2.2 Conditioning sets

Three conditioning sets are examined:

1. **Single-node:** $X = \{\texttt{AFF\_FEAR}\}$, set to +1 SD.
2. **Affect domain:** $X = \{\texttt{AFF\_FEAR}, \texttt{AFF\_WORRY}, \texttt{WORRY\_HEALTH\_SYSTEM}\}$, all set to +1 SD simultaneously.
3. **Behavior domain:** $X = \{\texttt{USE2\_MASK}, \texttt{USE2\_SPACE150}, \texttt{USE2\_HANDWASH20}, \texttt{USE2\_AVOID}\}$, all set to +1 SD simultaneously.

The primary outcome is $\Delta_{\texttt{SEVERITY}}$, the implied conditional displacement of the focal belief node. CMS is computed only for outcome nodes not included in the conditioning set.

Trust does not enter the conditioning operation; it defines which context-specific correlation matrix is used.

7.2.3 Network contexts

The analysis is applied identically to all six main-specification network contexts (waves 55–57 × low/high trust). Context enters only through the context-specific correlation matrix used in the CMS computation. Because the CMS operates on the full 8×8 Pearson correlation matrix rather than on the sparse GGM output, its reliability depends on the stability of correlation estimates; with sample sizes ranging from $n = 309$ to $n = 407$ across eight variables, the correlation matrix is well-determined (a conservative guideline of $n > 5p$ yields $n > 40$ here), so the CMS results are not expected to be materially affected by sampling instability in the correlation inputs.

7.2.4 Implementation

For each context, the CMS analysis uses the within-context Pearson correlation matrix R over the fixed node set as the Gaussian covariance approximation. Given conditioning indices X and outcome indices Y, the implied conditional displacement is computed in closed form as:

$$\Delta_Y = \Sigma_{YX}\Sigma_{XX}^{-1}x_{\text{shift}},$$

with $\Sigma = R$ and $x_{\text{shift}} = (c, \ldots, c)^\top$, where $c = +1$.

```
import numpy as np

def conditional_mean_shift(Sigma, x_idx, y_idx, c=1.0):
    """
    Sigma  : full correlation matrix (p x p numpy array)
    x_idx  : list of column indices for conditioning set X
    y_idx  : list of column indices for outcome set Y
    c      : shift magnitude in SD units (default +1)
    Returns: Delta_Y, implied displacement of Y nodes in SD units
    """
    Sigma_YX = Sigma[np.ix_(y_idx, x_idx)]
    Sigma_XX = Sigma[np.ix_(x_idx, x_idx)]
    x_shift  = np.full(len(x_idx), c)
    delta    = Sigma_YX @ np.linalg.solve(Sigma_XX, x_shift)
    return delta
```

Each variable corresponds to a fixed column index in Σ, used to specify `x_idx` and `y_idx` in the function call:

Index	Variable	Index	Variable
0	SEVERITY	4	USE2_MASK
1	AFF_FEAR	5	USE2_SPACE150
2	AFF_WORRY	6	USE2_HANDWASH20
3	WORRY_HEALTH_SYSTEM	7	USE2_AVOID

For the primary analysis, `y_idx = [0]` (i.e., `SEVERITY` is the sole outcome node), and `x_idx` is set to the column indices of the variables in the chosen conditioning set.

7.3 Results

7.3.1 Primary heatmap: $\Delta_{\texttt{SEVERITY}}$ across conditioning sets and contexts

Figure 7.1 shows the primary result: the implied conditional displacement of `SEVERITY` for each conditioning set (rows) and each network context (columns). Cell values are $\Delta_{\texttt{SEVERITY}}$ in SD units.

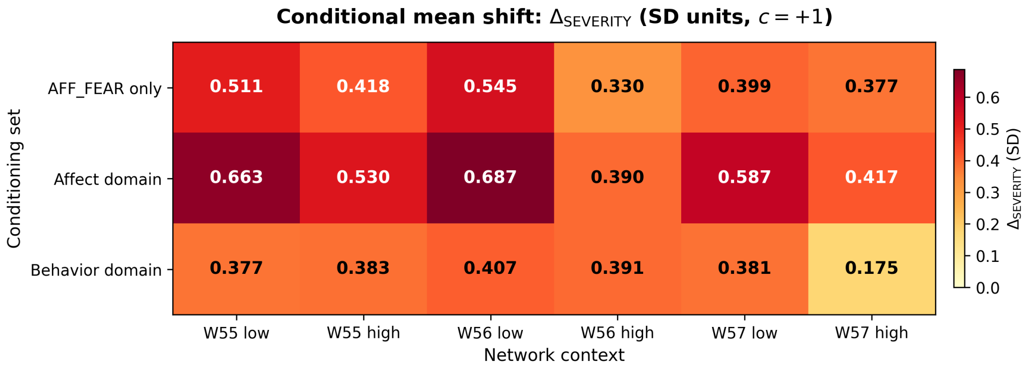

Figure 7.1: Conditional mean shift heatmap. Each cell shows the implied displacement $\Delta_{\texttt{SEVERITY}}$ (SD units) when the conditioning set is fixed to $c = +1$ SD. Rows: `AFF_FEAR` only (top), affect domain (middle), behavior domain (bottom). Columns: six main-specification network contexts. Color scale is zero-anchored; darker shading indicates larger implied displacement.

Table 7.1 reports the numerical values for $\Delta_{\texttt{SEVERITY}}$ per conditioning set and context.

Table 7.1: Implied conditional displacement of `SEVERITY` ($\Delta_{\texttt{SEVERITY}}$, SD units, $c = +1$) per conditioning set and network context.

Wave	Trust	AFF_FEAR only	Affect domain	Behavior domain
55	low	0.511	0.663	0.377
55	high	0.418	0.530	0.383
56	low	0.545	0.687	0.407
56	high	0.330	0.390	0.391
57	low	0.399	0.587	0.381
57	high	0.377	0.417	0.175

7.3.2 Key findings

Finding CMS-1: Affect-domain conditioning implies larger focal displacement than behavior-domain conditioning in most contexts. In five of six contexts, $\Delta_{\texttt{SEVERITY}}$ under affect-domain conditioning exceeded $\Delta_{\texttt{SEVERITY}}$ under behavior-domain conditioning. Wave 56 high-trust is the exception: behavior-domain conditioning produced a marginally larger displacement (0.391) than affect-domain conditioning (0.390), consistent with the lower P_{aff} observed for that context. This pattern extends the static finding that affect coupling generally dominates behavior coupling to the distributional geometry of the estimated joint model: in most contexts, conditioning on the affect domain implies a larger associational displacement in the focal belief than conditioning on the behavior domain under identical magnitude shifts ($c = +1$ SD).

Finding CMS-2: Low-trust contexts generally show larger affect-domain displacements than high-trust contexts. For affect-domain conditioning, $\Delta_{\texttt{SEVERITY}}$ was consistently larger in low-trust networks than in the corresponding high-trust networks across all three waves (0.663 vs. 0.530 at wave 55; 0.687 vs. 0.390 at wave 56; 0.587 vs. 0.417 at wave 57). For behavior-domain conditioning, the pattern is less consistent: wave 55 high-trust shows a slightly larger displacement (0.383) than low-trust (0.377). The affect-domain pattern is consistent with the higher global strength observed in low-trust networks: denser coupling implies larger associational displacement under affect-domain conditioning. This consistency between static embeddedness metrics and implied distributional displacement provides an internal validation of the embeddedness framework.

Finding CMS-3: High-trust contexts show the smallest displacements, with the specific context depending on the conditioning set. The two high-trust contexts with the lowest global strength — wave 56 high-trust ($GS = 2.656$) and wave 57 high-trust ($GS = 2.303$) — produced the smallest displacements, though which one ranked lowest depended on the conditioning set. For affect-domain and single-node conditioning, wave 56 high-trust showed the smallest displacement (affect domain: 0.390; `AFF_FEAR` only: 0.330). For behavior-domain conditioning, wave 57 high-trust showed the smallest displacement (0.175), consistent with the particularly limited behavioral coupling of `SEVERITY` in that context (only `USE2_HANDWASH20` retains a nonzero edge, with $|\hat{w}| = 0.041$). In both cases, the pattern is consistent with the general observation that weaker overall coupling implies smaller conditional displacements.

7.3.3 Secondary analysis: full displacement vectors

Figure 7.2 shows the full displacement vector Δ across all eight nodes under affect-domain conditioning for wave 55 low-trust versus wave 55 high-trust. This illustrates how the implied associational displacement propagates across the entire evaluative system, not merely at the focal belief node, and gives a direct visual sense of how the more densely coupled low-trust network implies larger displacements across most nodes.

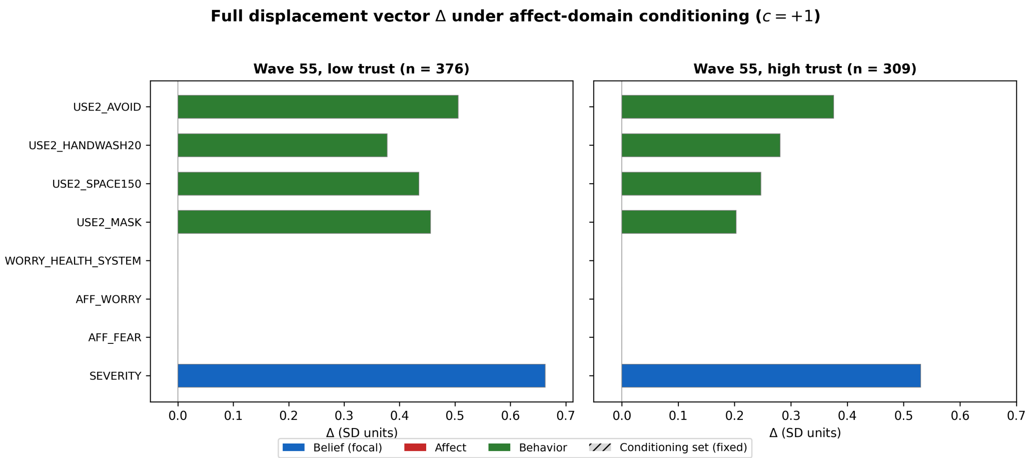

Figure 7.2: Full displacement vector Δ (SD units) across all eight nodes under affect-domain conditioning ($c = +1$ SD) for wave 55 low-trust (left, $n = 376$) versus wave 55 high-trust (right, $n = 309$). Node ordering: belief (blue), affect (red), behavior (green). Affect nodes have zero displacement by construction (they are the conditioning set). Larger displacements for the focal belief and most behavior nodes in the low-trust panel reflect higher cross-domain connectivity in that network.

7.3.4 Limitations of this analysis

Several limitations apply specifically to the conditional mean shift analysis. First, the conditional mean shift is a property of the estimated Gaussian joint distribution and does not constitute a simulation of belief change or a causal intervention. The implied displacements are associational statements about the modeled covariance structure and inherit all limitations of the GGM approximation (Section 8.5). Second, the analysis uses the point estimate of Σ without propagating estimation uncertainty; bootstrap confidence intervals on $\Delta_{\texttt{SEVERITY}}$ would be a natural extension for future work. Third, because the GGM is estimated on standardized variables, displacement magnitudes are in SD units relative to the within-context distribution and are not directly comparable across contexts with different marginal variances. Fourth, for wave 57 high-trust — the context with the sparsest estimated network (17 edges, $GS = 2.303$; Section 6.2) — `SEVERITY` has only one surviving behavior edge (`USE2_HANDWASH20`, $|\hat{w}| = 0.041$). As a result, the behavior-domain displacement of `SEVERITY` in that context ($\Delta_{\texttt{SEVERITY}} = 0.175$; Table 7.1) is driven almost entirely by this single weak edge rather than by a coherent behavior-domain coupling. This value should be interpreted with corresponding caution. More generally, for contexts where `SEVERITY` has limited direct edges to behavior nodes, the behavior-domain displacement relies partly on indirect paths through the affect domain; small non-zero off-diagonal covariance entries can produce small but non-trivially interpretable displacement values.

7.4 Code availability and interactive tool

The complete analysis pipeline — including preprocessing, network estimation, embeddedness metric computation, belief knowledge graph encoding, and conditional mean shift analysis — is openly available and reproducible. The code and data pipeline are hosted at:

https://github.com/imarouani/Thesis_Belief_Modeling

An interactive tool for exploring the conditional mean shift analysis across all nodes, conditioning sets, and estimation contexts is deployed at:

https://belief-modeling.streamlit.app/

The interactive tool allows the user to select any variable as the outcome node, any conditioning set, and any estimation context, and returns the implied displacement vector in real time. This provides an accessible interface to the GGM-based conditional geometry beyond what the static heatmap figures convey.

8 Discussion

This thesis investigated whether the structural coupling of a focal belief can be represented and quantified using an undirected conditional-dependence network model, stored as context-indexed objects in a belief knowledge graph, and illustrated computationally via a conditional mean shift analysis. The empirical analysis focused on a specific focal belief: respondents' belief about the severity of COVID-19, operationalized through the variable `SEVERITY`. The surrounding evaluative configuration included affective responses (fear and worry) and protective behavioral tendencies. Using COSMO survey data, networks were estimated separately across survey waves and trust-defined strata.

This chapter interprets the descriptive findings from Chapters 6 and 7 in relation to theoretical perspectives on belief systems and network psychometrics, discusses methodological implications, and outlines limitations.

8.1 Interpretation of the focal belief structure

Across all estimated network instances, the focal belief about the severity of COVID-19 was most strongly connected to affective variables rather than behavioral variables. Specifically, the strongest neighbor of `SEVERITY` was consistently either `AFF_FEAR` or `AFF_WORRY`. Behavioral variables, such as mask wearing or avoidance behaviors, were also conditionally associated with the focal belief but typically exhibited smaller edge weights.

This pattern suggests that the belief about the severity of COVID-19 was embedded primarily within an affective appraisal structure rather than directly within a behavioral action structure. Within the evaluated node set, the belief about the severity of COVID-19 therefore appears most tightly coupled to emotional responses.

One possible account, which cannot be tested within the present design, is that severity is fundamentally a threat appraisal, and fear and worry are the immediate emotional responses to perceived threat. They are psychologically close to the belief itself — part of the same appraisal process. Protective behaviors like mask-wearing or distancing, by contrast, sit further downstream. Acting on a severity belief requires not only perceiving the threat but also deciding to respond, which involves additional factors not captured in this node set: perceived efficacy, social norms, habit, convenience. The weaker conditional coupling between severity and behavior is therefore consistent with behavior being driven by a broader set of influences, only some of which pass through the severity belief. This reasoning is speculative, but it illustrates the kind of substantive question that the network representation makes visible.

This observation is compatible with the structural predictions of classical attitude frameworks that conceptualize attitudes as configurations consisting of cognitive, affective, and behavioral components (Breckler, 1984; Rosenberg & Hovland, 1960), with affect often playing a central role in the evaluative system. It is also compatible with the structural predictions of the Causal Attitude Network framework, which conceptualizes attitudes as networks of mutually reinforcing evaluative reactions (Dalege et al., 2016, 2019). Within CAN's consistency-pressure framework, this pattern is qualitatively expected: affective responses such as fear and worry function as the primary valence signals through which a threat appraisal is integrated and sustained across the evaluative system (Dalege et al., 2016). Nodes subject to stronger consistency pressure are predicted to show stronger mutual interdependencies, and affect—as the domain most closely associated with perceived threat severity—would plausibly show the strongest conditional coupling with the focal belief. However, as noted in Section 2.6.4, these results are consistent with CAN's structural predictions but do not confirm its causal claims.

This points to the broader value of the network approach for studying belief systems. Instead of collapsing the focal belief and its correlates into a latent summary, the network model preserves which components are directly linked to which. This makes it possible to examine how the focal belief is structurally embedded within the surrounding evaluative configuration.

8.2 Trust-context differences in belief connectivity

A second descriptive pattern concerned differences between trust-defined contexts. Networks estimated within the low-trust stratum exhibited higher global strength than networks estimated within the high-trust stratum in all three waves. Focal strength showed a similar directional pattern in waves 56 and 57, but was higher in the high-trust stratum in wave 55, suggesting that the trust-stratum contrast in local belief coupling is not uniform and varies by temporal context.

These patterns indicate that the belief–affect–behavior variables included in the analysis showed higher summed partial-correlation magnitudes among respondents reporting lower trust in the Robert Koch Institute at the level of global connectivity. One possible interpretation is that respondents with lower institutional trust lack an external anchor for their threat appraisal. High-trust individuals may partially offload the evaluation to the institution — treating institutional reassurance as a proxy for personal threat assessment — which could weaken the mutual dependencies among personal beliefs, emotions, and behaviors. Low-trust individuals, lacking that anchor, may rely more heavily on their own internal evaluative system, producing tighter coupling among its components. An alternative interpretation is that the stronger connectivity reflects response-style differences (more extreme or more internally consistent responding among low-trust respondents) rather than genuine psychological organization. The present design cannot distinguish between these accounts.

These differences are descriptive: the data are cross-sectional and the networks are estimated separately per subgroup, so they should not be read as causal effects of trust on belief structure. As noted in Section 3.5, the extreme-groups design means the magnitude of these differences is likely an overestimate relative to what would be found in a full-sample analysis.

8.3 Implications of the network representation

The network approach makes something visible that latent-variable models leave implicit: the specific pairwise structure linking a focal belief to its surrounding evaluative components. Rather than collapsing severity, fear, worry, and protective behaviors into a single factor score, the GGM preserves which components are directly linked to which, and how strongly. Focal strength and global strength then provide compact summaries of this structure that can be compared across contexts.

The conditional mean shift analysis introduced in Chapter 7 extends these static summaries to a distributional statement. It shows that the same structural differences captured by embeddedness metrics have measurable consequences for the implied conditional distribution: in more densely coupled networks, conditioning on the affect domain implies larger displacements of the focal belief. Moreover, the full displacement vector analysis (Figure 7.2) shows that this effect propagates system-wide — the low-trust network produces larger displacements not only at the focal node but across all non-conditioning nodes simultaneously, reflecting the overall higher connectivity of the system. This connection between static structure and distributional geometry confirms that embeddedness metrics index a meaningful property of the estimated model, not merely a descriptive label.

8.4 Contribution of the belief knowledge graph representation

Beyond the statistical estimation of belief networks, this thesis introduces a representation layer that stores each estimated network instance as an object within a belief knowledge graph. The purpose of this layer is not to modify the statistical model but to organize its outputs in a structured and queryable format.

Standard network outputs—figures and adjacency matrices—are well suited for inspecting a single network but do not scale as a representation standard when multiple instances must be compared across contexts. They lack built-in support for multi-instance storage under a uniform schema, explicit context indexing, attaching estimator settings and preprocessing notes, and running structured comparison queries across instances. By storing network instances within a property-graph structure, the BKG addresses each of these gaps. This is not a claim that property-graph representations are universally superior to alternative storage formats; rather, the BKG is especially well suited here because the thesis requires storing repeated network instances, stable variable identities across instances, context-specific node metrics, edge-level statistics, provenance and estimation settings, and cross-instance retrieval and comparison—requirements that the typed relational structure of a property graph handles naturally.

Within this representation, each estimated network corresponds to a specific estimation context defined by survey wave and trust stratum. Nodes represent variables, edges represent estimated conditional associations, and additional properties store metrics such as focal strength, global strength, and bootstrap-based stability summaries. Queries can then retrieve structural properties of the belief networks while maintaining explicit links to the estimation context and modeling parameters. Example Cypher queries demonstrating these retrieval operations are provided in Section 5.5.

Compared to flat CSV files or relational table storage, the property-graph representation offers graph traversal queries, semantic typing of relationships, and native multi-hop retrieval without custom join logic. Compared to a relational database, the property-graph structure avoids the need to define rigid schemas before insertion and accommodates heterogeneous edge properties naturally. In all cases, the contribution is the demonstration that belief-network outputs can be transformed from static analytical artifacts into structured objects that support transparent comparison and reproducible analysis. The representation improves auditability, provenance tracking, transparent retrieval, and inspectable relation structure—properties that are relevant to broader goals of explainable, trustworthy, and transparent analytical systems, though a full implementation of such explainability goals remains a direction for future work.

8.5 Limitations

The study has several limitations.

First, as noted in Section 1.5, the analysis uses repeated cross-sectional data in which respondents are not tracked across waves; the results therefore describe differences between estimation contexts rather than longitudinal belief change, and no within-person inferences are warranted.

Second, the node set used in the analysis is intentionally minimal, consisting of a single focal belief about the severity of COVID-19 together with a small set of affective and behavioral variables. While this design supports a clear proof of concept, real-world belief systems are influenced by a broader range of cognitive, social, and informational factors. With only one belief node, the "structure" observed in the networks is more precisely described as focal-belief coupling than belief structure in the broader sense.

Third, the network estimation relies on a Gaussian graphical model applied to ordinal Likert-type items treated as numeric variables. Although this approximation is common in exploratory network analysis and is generally considered acceptable for items with five or more ordered categories (Rhemtulla et al., 2012), the modeling choice remains an approximation of the underlying response structure. A Spearman rank-correlation robustness check was conducted for all six main-specification contexts (Section 6.8.3) and confirmed that the main structural findings are preserved under an alternative correlation input (top-neighbor match in five of six contexts; $r_s = 0.886$, $p = .019$ for focal strength orderings). Polychoric correlations under a latent-Gaussian assumption were not assessed and remain a possible extension for future work.

Fourth, the comparison of network structures across contexts is descriptive rather than inferential. Formal statistical tests for network differences, such as the Network

Comparison Test (van Borkulo et al., 2023), were not implemented. Given the unequal sample sizes across trust strata, the temporal confound between wave and trust differences, and the scope of the thesis, such inferential comparison would require a dedicated study design with matched subsamples and controlled temporal confounds. The conditional mean shift analysis introduced in Chapter 7 provides a complementary computational illustration of structural differences without requiring inferential claims about population-level network differences.

Fifth, the resampling-based stability summaries (`prop_nonzero`, bootstrap percentile intervals) reflect joint variability arising from both sampling variation and regularization. Because EBICglasso regularization can set edges to zero, `prop_nonzero` captures both true edge absence and regularization-induced sparsification in bootstrap samples. These summaries are therefore exploratory heuristics rather than classical confidence coverage statements, and should be interpreted accordingly.

Sixth, comparisons of focal strength $S(\texttt{SEVERITY})$ and global strength GS across network instances should be interpreted cautiously. EBICglasso selects different sparsity levels depending on sample size, correlation structure, and the selected regularization path, which means the sum of absolute edge weights is not directly comparable across estimation contexts without accounting for differences in the number and magnitude of retained edges. The descriptive differences reported should be understood with this caveat in mind.

Seventh, case-dropping stability (CS-coefficient; Epskamp et al., 2018) was acceptable (≥ 0.50) in five of six contexts (Table 6.6). The single exception was wave 55 high-trust (CS = 0.440), which fell modestly below the conventional threshold. Strength-based embeddedness claims for that context should therefore be treated with additional caution (see Section 6.8 for full details).

Eighth, as detailed in Section 3.5, the extreme-groups design inflates observed low- vs. high-trust differences; findings should be understood as upper-bound characterizations rather than population-representative estimates (Preacher et al., 2005). Additionally, trust strata defined by a single institutional trust item may co-vary with unmeasured third variables — including education, political orientation, or general conspiratorial reasoning — so observed between-stratum differences in network structure cannot be unambiguously attributed to institutional trust itself.

Ninth, the wave 57 high-trust network shows the lowest global strength and fewest edges among the six main-specification contexts ($GS = 2.303$, 17 edges). Under the more conservative thresholded specification, only 6 edges survive for this context, and a single bridge edge remains. The substantive and methodological explanations for this pattern — genuine psychological decoupling versus ceiling-effect attenuation in behavioral items — cannot be distinguished in the present design. Cross-context comparisons involving this network instance should therefore be interpreted with awareness of this ambiguity. A specific consequence for the conditional mean shift analysis in this context is discussed in Section 7.3.4.

These limitations reflect the proof-of-concept nature of the present work and define the boundaries within which the results should be interpreted.

8.6 Future directions

The present work opens several directions for further work.

First, the framework could be extended from a single focal-belief network to multiple overlapping networks with different focal nodes. The current node set contains one belief variable (SEVERITY), but a richer set might include perceived personal risk, perceived efficacy of protective measures, or trust in scientific institutions. With multiple belief nodes, the pipeline could be run with each belief as focal in turn, producing a family of focal-belief configurations stored in the same BKG. The same variable would then appear as the focal node in one configuration and as a peripheral node in another — and its embeddedness metrics would differ across these roles.

This matters because beliefs are not all at the same level. Some beliefs are specific and concrete (“COVID-19 is severe”), others are broader evaluations (“the government is handling this well”), and others sit at an even more abstract level (“science is trustworthy”). A higher-order belief like institutional trust may shape whether someone endorses a lower-order belief like perceived severity, which in turn shapes affective and behavioral responses. A single flat network cannot express this kind of structure — all nodes sit at the same level. But the BKG can, because its schema separates what a variable *is* (its identity as a Variable node) from what it *does* in a specific network (its role, strength, and neighbors as a NodeInstance). If the same variable shows high embeddedness when focal but low embeddedness when peripheral, that asymmetry is itself informative about the variable’s structural position in the belief system. Querying the BKG across multiple focal-belief configurations would make these hierarchical patterns visible — not as a tree imposed from theory, but as an emergent property of the estimated network structures. The current thesis demonstrates the pipeline for one focal belief; the BKG provides the representational foundation for scaling it to many.

Second, the conditional mean shift analysis introduced in Chapter 7 is limited by the associational constraints of the cross-sectional GGM. With longitudinal panel data and directed models — for instance, constraint-based causal discovery methods — the same framework could support formal intervention analysis rather than the associational displacement approach used here. That would be a natural step toward computational modeling of belief change.

Third, methodological robustness could be extended further. Polychoric correlations under a latent-Gaussian assumption remain untested as an alternative correlation input. Formal network comparison procedures such as the Network Comparison Test (van Borkulo et al., 2023) could be applied under study designs better suited for inferential comparison than the present extreme-groups approach.

Fourth, the analytical approach developed here is not specific to COVID-19 beliefs or the German population. Applying the pipeline to other survey data — political attitudes, health behaviors, cross-cultural comparisons — would test the generalizability of the embeddedness framework and the conditional mean shift interpretation.

These directions represent extensions of the present workflow, not achievements of the current thesis.

8.7 Conclusion

The findings support the central methodological claim of this thesis: a focal belief can be represented as part of an explicit belief–affect–behavior network, compared across substantively defined estimation contexts using connectivity-based summaries, stored and queried in a property-graph architecture, and examined computationally via a conditional mean shift analysis.

The estimated networks showed that perceived COVID-19 severity was consistently coupled most strongly to affective variables across all six contexts. Low-trust networks exhibited higher global connectivity in every wave. These patterns held across all four robustness dimensions — regularization strength, post-selection thresholding, Spearman correlation input, and resampling-based stability diagnostics — as documented in Section 6.8. The conditional mean shift analysis confirmed that the affect domain implied larger focal-belief displacements than the behavior domain in most contexts, and that low-trust networks generally implied larger displacements, providing an internal consistency check between static embeddedness metrics and distributional properties of the estimated model.

At the same time, the scope remains deliberately limited. The results are descriptive, context-specific, and based on a small fixed node set. The thesis should therefore be read as a proof of concept for structured representation of focal-belief coupling, not as a definitive account of belief change or belief-system dynamics.

Appendix

A. COSMO Codebook Details for Thesis Variables

Source: COSMO codebook (`codebook_COSMO.csv`). Table 1 lists the codebook information for variables used in the thesis across waves 55–57.

Table 1: Codebook information for variables used in the thesis (waves 55–57).

Variable	**Codebook entry (EN, condensed)**
`SEVERITY`	Perceived severity of COVID-19. Scale: 1 = totally harmless; ...; 7 = extremely dangerous.
`AFF_FEAR`	Affect item (fear). Raw endpoints: scary – not scary. Reverse-coded in analysis so that higher values reflect higher fear.
`AFF_WORRY`	Affect item (worry). Raw endpoints: worrying – not worrying. Reverse-coded in analysis so that higher values reflect higher worry.
`WORRY_HEALTH_SYSTEM`	Worry about the health system. Scale: 1 = very few worries; ...; 7 = very many worries. Not reverse-coded (higher values already indicate more worry).
`USE2_MASK`	Protective behavior frequency. Scale: 1 = Trifft nicht zu (recoded to NA); 2 = Nie; 3 = Selten; 4 = Manchmal; 5 = Häufig; 6 = Immer. Effective range after recoding: 2–6.
`USE2_SPACE150`	Protective behavior frequency. Scale: 1 = Trifft nicht zu (recoded to NA); 2 = Nie; ...; 6 = Immer. Effective range after recoding: 2–6.
`USE2_HANDWASH20`	Protective behavior frequency. Scale: 1 = Trifft nicht zu (recoded to NA); 2 = Nie; ...; 6 = Immer. Effective range after recoding: 2–6.
`USE2_AVOID`	Protective behavior frequency. Scale: 1 = Trifft nicht zu (recoded to NA); 2 = Nie; ...; 6 = Immer. Effective range after recoding: 2–6.

Variable	Codebook entry (EN, condensed)
TRUST_RKI	Trust in the Robert Koch Institute (RKI). Categories 1 (*keine Angabe möglich*; no answer possible) and 2 (non-substantive; substantive range begins at 3) recoded to NA; higher values indicate higher trust. Effective range after recoding: 3–9. Used only as a context-stratification variable; not included as a node in the belief–affect–behavior network.

B. Trust Stratum Cut-Points and Sample Sizes

Table 2 reports the within-wave TRUST_RKI tercile cut-point values and complete-case sample sizes for each network instance. Cut-points are expressed in the original TRUST_RKI scale units and were computed using strict within-wave terciles as described in Section 3.5.

Table 2: Within-wave TRUST_RKI tercile boundaries and complete-case sample sizes per network instance. Boundaries (q_{33}, q_{67}) are computed on non-missing TRUST_RKI values within each wave. *Low*: respondents with TRUST_RKI $\leq q_{33}$; *High*: respondents with TRUST_RKI $\geq q_{67}$; middle tercile excluded.

Wave	Trust stratum	n_{cc}	q_{33}, q_{67}	Assignment rule
55	low	376	6, 8	$\leq q_{33}$
55	high	309		$\geq q_{67}$
56	low	383	6, 8	$\leq q_{33}$
56	high	351		$\geq q_{67}$
57	low	336	6, 8	$\leq q_{33}$
57	high	407		$\geq q_{67}$

Note. Cut-point values are on the original TRUST_RKI scale (effective range 3–9 after recoding). All three waves yielded the same boundaries: low trust ≤ 6, high trust ≥ 8, middle tercile (7) excluded.

C. Belief Knowledge Graph: Additional Cypher Query Examples

The following queries extend the three examples provided in Section 5.5. All queries assume the BKG has been imported into Neo4j using the provided `import_kg.cypher` script.

Query 4 — Identify contexts where `SEVERITY` has no direct behavior connections:

```
MATCH (net:NetworkInstance)-[:HAS_CONTEXT]->(ctx:Context)
WHERE NOT EXISTS {
  MATCH (net)-[:HAS_EDGE_INSTANCE]->(ei:EdgeInstance),
        (ei)-[:FROM|TO]->
             (ni:NodeInstance)-[:INSTANCE_OF]->(v:Variable {role:"
    focal_belief"}),
        (ei)-[:FROM|TO]->
             (ni2:NodeInstance)-[:INSTANCE_OF]->(v2:Variable {domain:"
    behavior"})
  WHERE id(v) <> id(v2)
}
RETURN ctx.wave, ctx.trust_stratum;
```

Result: No rows returned. In all six main-specification contexts, `SEVERITY` retains at least one direct conditional association with a behavioral variable.

Query 5 — Identify `SEVERITY`'s strongest behavior neighbor per context:

```
MATCH (ei:EdgeInstance),
      (ei)-[:FROM|TO]->
           (n1:NodeInstance)-[:INSTANCE_OF]->(v1:Variable {role:"
    focal_belief"}),
      (ei)-[:FROM|TO]->
           (n2:NodeInstance)-[:INSTANCE_OF]->(v2:Variable {domain:"
    behavior"}),
      (net:NetworkInstance)-[:HAS_EDGE_INSTANCE]->(ei),
      (net)-[:HAS_CONTEXT]->(ctx:Context)
WHERE v1 <> v2
WITH ctx.wave AS wave, ctx.trust_stratum AS trust,
     v2.node_id AS behavior_node, ei.abs_weight AS w
ORDER BY wave, trust, w DESC
WITH wave, trust, COLLECT({node: behavior_node, weight: w})[0] AS top
RETURN wave, trust,
```

```
      top.node   AS strongest_behavior_neighbor,
      top.weight AS abs_weight
ORDER BY wave, trust;
```

Result (main specification, $\gamma = 0.5$):

wave	trust	strongest_behavior_neighbor	abs_weight
55	high	USE2_AVOID	0.190
55	low	USE2_MASK	0.092
56	high	USE2_AVOID	0.246
56	low	USE2_AVOID	0.176
57	high	USE2_HANDWASH20	0.041
57	low	USE2_AVOID	0.109

These values correspond to the behavioral-neighbor findings reported in Section 6.2 and Finding 2 in Chapter 6, confirming that the BKG stores and retrieves estimation outputs without transformation.

D. Behavioral Item Distributions by Trust Stratum and Wave

Table 3 reports the mean and standard deviation of each behavioral item within each trust stratum across waves 55–57. These distributional statistics support the methodological interpretation discussed in Section 6.2: consistently higher means and lower standard deviations in the high-trust stratum, particularly in wave 57, are consistent with ceiling effects that would attenuate Pearson correlations and drive EBICglasso toward sparser solutions.

Table 3: Mean (SD) of behavioral items within each trust stratum and wave. Effective response range: 2–6 after recoding (1 = Trifft nicht zu recoded to NA). Higher means approaching 6 with lower SDs indicate ceiling-effect compression.

Wave	Trust	USE2_MASK		USE2_SPACE150		USE2_HANDWASH20		USE2_AVOID	
		M	SD	M	SD	M	SD	M	SD
55	low	5.16	1.04	4.69	1.13	4.74	1.22	3.95	1.30
55	high	5.68	0.53	5.28	0.77	5.14	0.95	4.35	1.12
56	low	5.39	0.88	4.85	1.03	4.93	1.07	4.19	1.26
56	high	5.67	0.59	5.22	0.80	5.08	1.01	4.51	1.05
57	low	5.26	1.03	4.84	1.07	4.78	1.21	4.27	1.24
57	high	5.73	0.51	5.35	0.67	5.21	0.87	4.77	0.98

Note. Values computed from complete-case instance files (effective range 2–6 after recoding 1 = Trifft nicht zu to NA). High-trust strata show consistently higher means and lower standard deviations across all behavioral items, consistent with ceiling effects that may attenuate Pearson correlations in those contexts (see Section 4.1.1).

References

Betsch, C., et al. (2024). COVID-19 snapshot monitoring (COSMO germany) – data of 69 surveys on pandemic attitudes, feelings and behaviours. https://doi.org/10.23668/PSYCHARCHIVES.15213

Betsch, C., Eitze, S., Korn, L., Felgendreff, L., Tänzer, A.-S., Sprengholz, P., et al. (2025). Dataset of 69 german COVID-19 snapshot monitoring (COSMO) surveys on pandemic attitudes, feelings and behaviours. *Scientific Data*, *12*, Article 1066. https://doi.org/10.1038/s41597-025-05275-3

Borsboom, D., & Cramer, A. O. J. (2013). Network analysis: An integrative approach to the structure of psychopathology. *Annual Review of Clinical Psychology*, *9*, 91–121. https://doi.org/10.1146/annurev-clinpsy-050212-185608

Borsboom, D., Deserno, M. K., Rhemtulla, M., Epskamp, S., Fried, E. I., McNally, R. J., Robinaugh, D. J., Perugini, M., Dalege, J., Costantini, G., Isvoranu, A.-M., Wysocki, A. C., van Borkulo, C. D., van Bork, R., & Waldorp, L. J. (2021). Network analysis of multivariate data in psychological science. *Nature Reviews Methods Primers*, *1*, Article 58. https://doi.org/10.1038/s43586-021-00055-w

Borsboom, D., Mellenbergh, G. J., & van Heerden, J. (2003). The theoretical status of latent variables. *Psychological Review*, *110*(2), 203–219. https://doi.org/10.1037/0033-295X.110.2.203

Breckler, S. J. (1984). Empirical validation of affect, behavior, and cognition as distinct components of attitude. *Journal of Personality and Social Psychology*, *47*(6), 1191–1205. https://doi.org/10.1037/0022-3514.47.6.1191

Cohen, G. L. (2003). Party over policy: The dominating impact of group influence on political beliefs. *Journal of Personality and Social Psychology*, *85*(5), 808–822. https://doi.org/10.1037/0022-3514.85.5.808

Dalege, J., Borsboom, D., van Harreveld, F., van den Berg, H., Conner, M., & van der Maas, H. L. J. (2016). Toward a formalized account of attitudes: The Causal Attitude Network (CAN) model. *Psychological Review*, *123*(1), 2–22. https://doi.org/10.1037/a0039802

Dalege, J., Borsboom, D., van Harreveld, F., & van der Maas, H. L. J. (2019). The attitudinal entropy (AE) framework as a general theory of individual attitudes. *Psychological Inquiry*, *29*(4), 175–193. https://doi.org/10.1080/1047840X.2018.1537246

Ecker, U. K. H., Lewandowsky, S., Cook, J., Schmid, P., Fazio, L. K., Brashier, N., et al. (2022). The psychological drivers of misinformation belief and its resistance to correction. *Nature Reviews Psychology*, *1*, 13–29. https://doi.org/10.1038/s44159-021-00006-y

Epskamp, S., Borsboom, D., & Fried, E. I. (2018). Estimating psychological networks and their accuracy: A tutorial paper. *Behavior Research Methods*, *50*, 195–212. https://doi.org/10.3758/s13428-017-0862-1

Epskamp, S., Cramer, A. O. J., Waldorp, L. J., Schmittmann, V. D., & Borsboom, D. (2012). Qgraph: Network visualizations of relationships in psychometric data. *Journal of Statistical Software*, *48*(4), 1–18. https://doi.org/10.18637/jss.v048.i04

Festinger, L. (1957). *A theory of cognitive dissonance*. Stanford University Press.

Flora, D. B., & Curran, P. J. (2004). An empirical evaluation of alternative methods of estimation for confirmatory factor analysis with ordinal data. *Psychological Methods*, *9*(4), 466–491. https://doi.org/10.1037/1082-989X.9.4.466

Foygel, R., & Drton, M. (2010). Extended bayesian information criteria for gaussian graphical models. *Advances in Neural Information Processing Systems*, *23*.

Friedman, J., Hastie, T., & Tibshirani, R. (2008). Sparse inverse covariance estimation with the graphical lasso. *Biostatistics*, *9*(3), 432–441. https://doi.org/10.1093/biostatistics/kxm045

Hasher, L., Goldstein, D., & Toppino, T. (1977). Frequency and the conference of referential validity. *Journal of Verbal Learning and Verbal Behavior*, *16*(1), 107–112. https://doi.org/10.1016/S0022-5371(77)80012-1

Hogan, A., Blomqvist, E., Cochez, M., d'Amato, C., Melo, G. d., Gutierrez, C., Kirrane, S., Labra Gayo, J. E., Navigli, R., Neumaier, S., Ngomo, A.-C. N., Polleres, A., Rashid, S. M., Rula, A., Schmelzeisen, L., Sequeda, J., Staab, S., & Zimmermann, A. (2021). Knowledge graphs. *ACM Computing Surveys*, *54*(4), 1–37. https://doi.org/10.1145/3447772

Holgado-Tello, F. P., Chacón-Moscoso, S., Barbero-García, I., & Vila-Abad, E. (2010). Polychoric versus Pearson correlations in exploratory and confirmatory factor analysis of ordinal variables. *Quality & Quantity*, *44*(1), 153–166. https://doi.org/10.1007/s11135-008-9190-y

Hornsey, M. J. (2020). Why facts are not enough: Understanding and managing the motivated rejection of science. *Current Directions in Psychological Science*, *29*(6), 583–591. https://doi.org/10.1177/0963721420969364

Hovland, C. I., Janis, I. L., & Kelley, H. H. (1953). *Communication and persuasion: Psychological studies of opinion change*. Yale University Press.

Kahan, D. M. (2013). Ideology, motivated reasoning, and cognitive reflection. *Judgment and Decision Making*, *8*(4), 407–424.

Kahan, D. M., Jenkins-Smith, H., & Braman, D. (2011). Cultural cognition of scientific consensus. *Journal of Risk Research*, *14*(2), 147–174. https://doi.org/10.1080/13669877.2010.511246

Lauritzen, S. L. (1996). *Graphical models*. Oxford University Press.

Lewandowsky, S., Ecker, U. K. H., Seifert, C. M., Schwarz, N., & Cook, J. (2012). Misinformation and its correction: Continued influence and successful debiasing. *Psychological Science in the Public Interest*, *13*(3), 106–131. https://doi.org/10.1177/1529100612451018

MacKuen, M., Wolak, J., Keele, L., & Marcus, G. E. (2010). Civic engagements: Resolute partisanship or reflective deliberation. *American Journal of Political Science*, *54*(2), 440–458. https://doi.org/10.1111/j.1540-5907.2010.00440.x

Marcus, G. E., Neuman, W. R., & MacKuen, M. (2000). *Affective intelligence and political judgment*. University of Chicago Press.

Neo4j, Inc. (2024). Neo4j graph data platform [Accessed 2024]. https://neo4j.com

Newman, M. E. J. (2010). *Networks: An introduction*. Oxford University Press. https://doi.org/10.1093/acprof:oso/9780199206650.001.0001

Pennycook, G., Cannon, T. D., & Rand, D. G. (2018). Prior exposure increases perceived accuracy of fake news. *Journal of Experimental Psychology: General*, *147*(12), 1865–1880. https://doi.org/10.1037/xge0000465

Pornpitakpan, C. (2004). The persuasiveness of source credibility: A critical review of five decades' evidence. *Journal of Applied Social Psychology*, *34*(2), 243–281. https://doi.org/10.1111/j.1559-1816.2004.tb02547.x

Preacher, K. J., Rucker, D. D., MacCallum, R. C., & Nicewander, W. A. (2005). Use of the extreme groups approach: A critical reexamination and new recommendations. *Psychological Methods*, *10*(2), 178–192. https://doi.org/10.1037/1082-989X.10.2.178

Rhemtulla, M., Brosseau-Liard, P. E., & Savalei, V. (2012). When can categorical variables be treated as continuous? a comparison of robust continuous and categorical SEM estimation methods under suboptimal conditions. *Psychological Methods*, *17*(3), 354–373. https://doi.org/10.1037/a0029315

Rosenberg, M. J., & Hovland, C. I. (1960). Cognitive, affective, and behavioral components of attitudes. In M. J. Rosenberg, C. I. Hovland, W. J. McGuire, R. P. Abelson, & J. W. Brehm (Eds.), *Attitude organization and change* (pp. 1–14). Yale University Press.

Tajfel, H., & Turner, J. C. (1986). The social identity theory of intergroup behavior. In S. Worchel & W. G. Austin (Eds.), *Psychology of intergroup relations* (2nd ed., pp. 7–24). Nelson-Hall.

van Borkulo, C. D., van Bork, R., Boschloo, L., Kossakowski, J. J., Tio, P., Schoevers, R. A., Borsboom, D., & Waldorp, L. J. (2023). Comparing network structures on three aspects: A permutation test. *Psychological Methods*, *28*(6), 1273–1285. https://doi.org/10.1037/met0000476